EV HV Component EMC Test Plan 21 October 2022, Template v3.1

Product Name: Head UP Display (Gen 3)	eTracker/Jira No: EMC-4394
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Product Manager: Dong Li (li-dong@crystal-optech.com) Product Part Number(s): T8U2-14F906-BA T8U2-14F906-AA T2L2-14F906-BA T2L2-14F906-AA	Vehicles & Model Year using this product: L481 26MY L384 26MY
Product Manufacturing Location(s) *Building 9, Dongfeng Science and Technology Industrial Park, No. 4 Xingye Road, Songshan Lake Ecological Park, Dongguan City, Guangdong Province, China*	EMC Specification Used: JLR-EMC-CS v1.0 Amendment 4

I certify that the information contained in this test plan is factual including description of the product operation, correct functional classifications, and acceptance criteria. I understand and agree that any subsequent changes to this test plan prior to design verification testing shall be communicated to the JLR EMC department. Any changes or revisions to this test plan after test completion shall require written technical justification and approval by the same EMC department. I understand that failure to follow this process may result in non-acceptance of the product's EMC test data by the JLR EMC department. I also understand and acknowledge that requirements validated via this test plan are relevant only to the specific vehicles that the product is to be fitted to. Use of the product on other vehicle platforms may require additional EMC performance requirements, which will necessitate additional verification testing of the product. I certify that the product samples submitted for EMC testing are of a production representative design. I agree to submit a summary report directly to the JLR EMC department no later than five (5) business days following completion of testing. I also agree to forward a copy of the test laboratory's detailed test report directly to the JLR EMC department within thirty (30) business days following completion of testing.

Supplier Product Design Engineer:

Sign	Print	Date
陆鸿民 冯军伟	Lu Hongmin Feng Junwei	2024.3.29

JLR Component Owner Concurrence:

Sign	Print	Date
	RYAN HARRISON	08/04/2024

JLR EMC Concurrence

Sign	Print	Date
	Jarnail Bharaj	08/04/2024

Test Plan Revision History

Date	Supplier Reference	Description	Approved Issue No.
21/07/2023	*V0.1*	*JLR-EMC-2023-0134, First Draft*	
18/08/2023	*V0.2*	*JLR-EMC-2023-0134, Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv1.0 -JB review 31-07-23*	
07/09/2023	*V0.3*	*JLR-EMC-2023-0134, Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv0.2 -JB&RH review 21-08-2023*	
15/09/2023	*V0.4*	*JLR-EMC-2023-0134, Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv0.3_JLR Review 11092023*	
25/09/2023	*V0.5*	*JLR-EMC-2023-0134, Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv0.4_JLR Review 15092023*	
28/09/2023	*V0.6*	*JLR-EMC-2023-0134, Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv0.4_JLR Review 26092023*	
09/10/2023	*V0.7*	*JLR-EMC-2023-0134,Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv0.4_JLR Review 26092023*	
12/10/2023	*V0.8*	*JLR-EMC-2023-0134,Test plan was updated based on the JLR comments "EMC-4394 HUD 3 TPv0.4_JLR Review 26092023*	
29/03/2024	*V0.9*	*JLR-EMC-2023-0134,Test plan was updated : Modify the middle housing ground leg*	

For Internal EMC Department Use		
Test Plan Tracking Number	JLR-EMC-202 - 0134	Issue No. 2.0

Table of Contents

Abbreviations

AN

Artificial network

C1: Power supply interface (MX34007UF1)

C2: Video signal interface (E6S10A-40MT5-A)

CCCM

Computing, Communication, Control Module

DUT

Device(s) Under Test.

HUD

Head UP Display

LISN

line impedance stabilization network

LHD

Left Hand Drive

LCR

Inductance Capacitance Resistance

N/C

No connection

N/A

Not applicable

PC

Personal Computer

RHD

Right Hand Drive

1.0 Introduction

The test plan is made for components of Head UP Display (GEN 3) Project and can be used as a guide to the lab. This document includes contents of product introduction, ways of building setups, test process and test criteria. This test plan is in accordance with JLR-EMC-CS_1.0_AMENDMENT_4.

1.1 Product Family Description

L384 model HUD is divided into two projects according to the position of the steering wheel, namely T8U2-14F906-BA (LHD) and T8U2-14F906-AA (RHD). The HUDs of these two projects are mirror images in design. Similarly, L481 model also has two projects, namely T2L2-14F906-BA (LHD) and T2L2-14F906-AA (RHD), and the HUD designs with the same steering wheel position in these four projects are completely consistent.

In summary: T8U2-14F906-BA (LHD) equates to T2L2-14F906-BA (LHD) , T8U2-14F906-AA (RHD) equates to T2L2-14F906-AA (RHD); The HUDs of RHD and LHD are identical except for the structural mirror image.

1.2 Theory of Operation

The HUD is installed under the IP platform in front of the driver's seat in the car cockpit.It is directly powered by a battery (not connected to the ignition switch).Both the image signal and the enable signal of the HUD come from CCCM.

HUD can be started and turned off according to the driver's needs before or during driving.

When the HUD is turned off, the motor will rotate to its initial position, However, if the HUD power supply is abnormally disconnected (that is, when the enable line is normally connected, the power is abnormally disconnected), the motor will stay in this position because there is no power. This situation can be caused by non-driver operation.

When the HUD is turned on, the motor will return to the previous memory angle (It takes 2~4s from when the HUD starts to when the motor rotates to the memory point.).

The PGU unit is responsible for projecting driving information into the driver's field of vision through optical technology.The product puts the driver's perspective in the center and integrates display elements such as instrument information and navigation.Image information is projected onto the driver's line of sight through an optical system to enhance the driving experience.

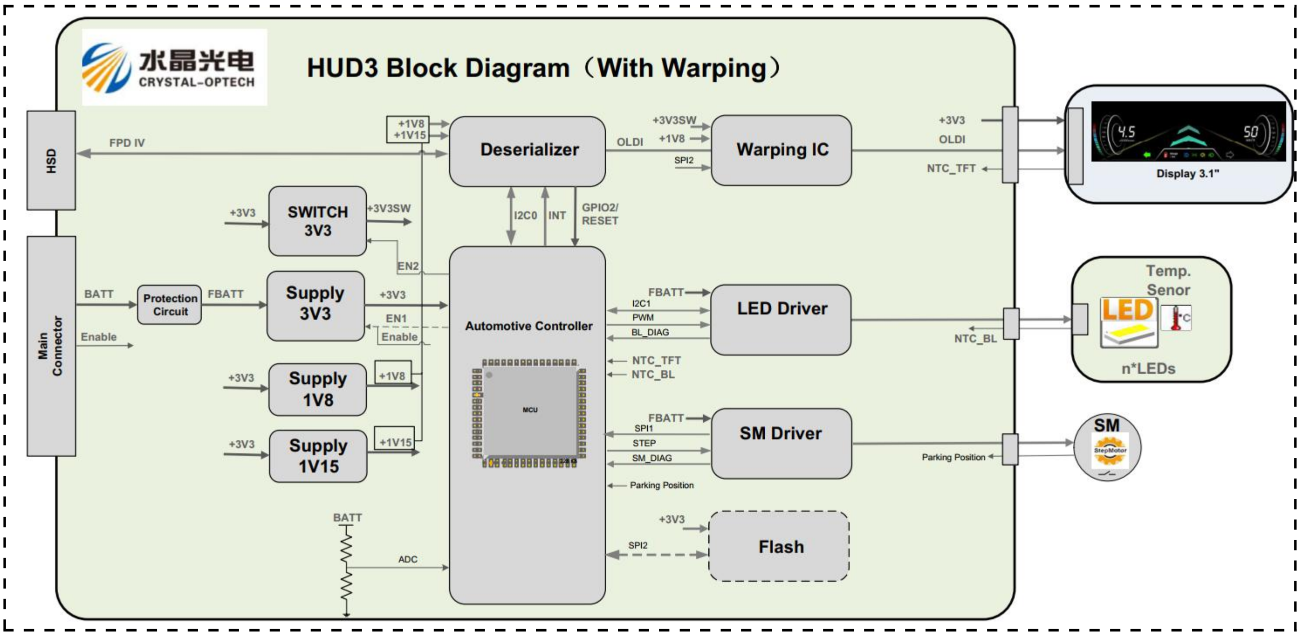


Figure 1-1: Internal block diagram of the HUD

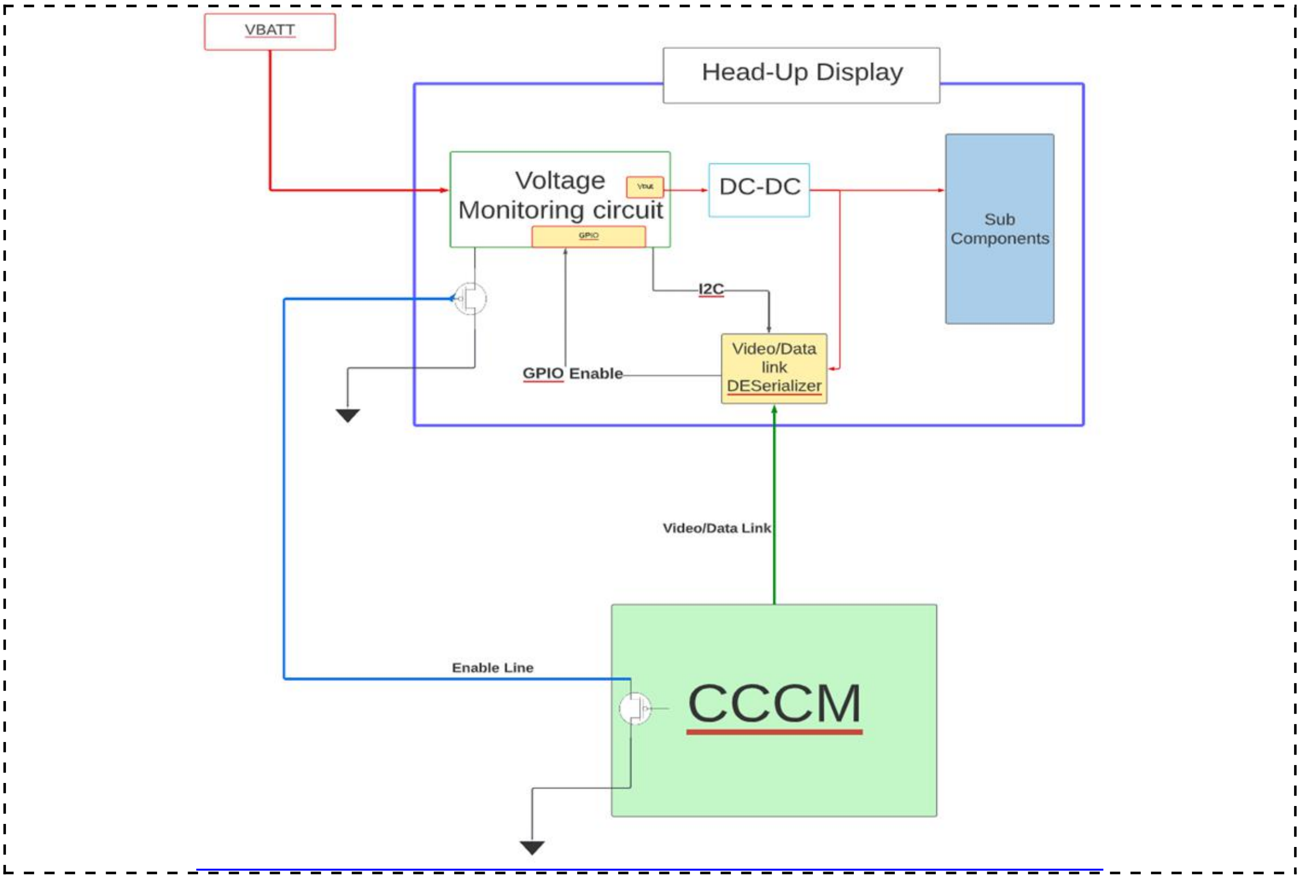


Figure 1-2: Sub-System Interfaces – Configuration

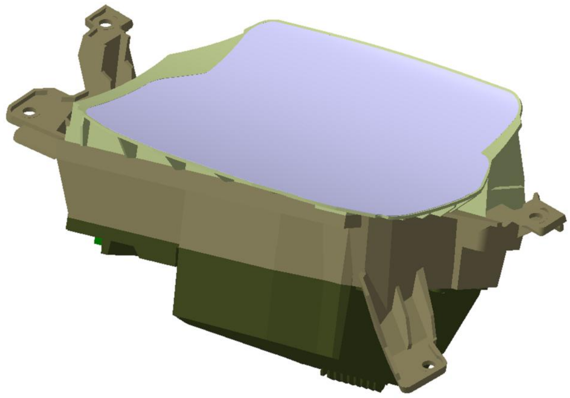
1.3 Physical Construction

a) What are the Product Package Material(s):							
Metallic ►	☑	Non-Metallic ►	☑	Conductive ►	☑	Non-Conductive ►	☑

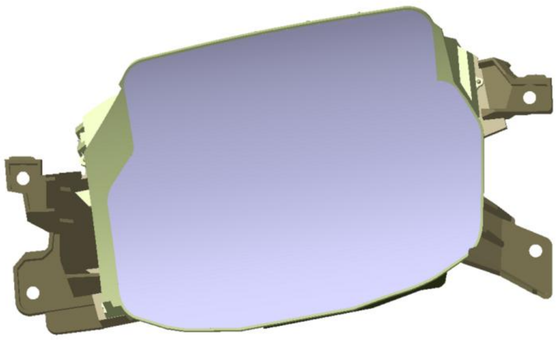
Provide further Material description below as necessary:
Non-Metallic/ Non-Conductive ： Cover Lens/ Top Housing
Metallic/ Conductive ： Middle Housing /Bottom Housing

b) What is the Product Package Volume:					
X Dimension (mm) ►	428	Y Dimension (mm) ►	255	Z Dimension (mm) ►	250

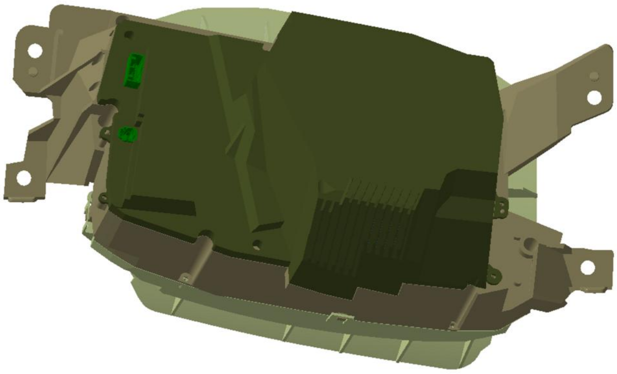
c) Provide the Product Mechanical View:



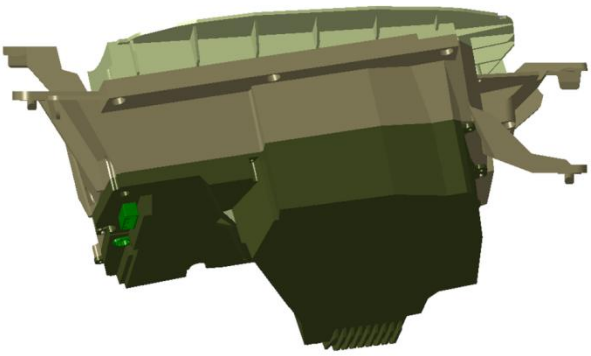
ISO View



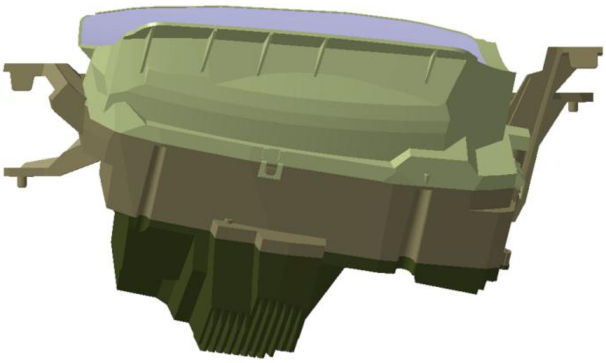
Top – Plan View



Bottom View

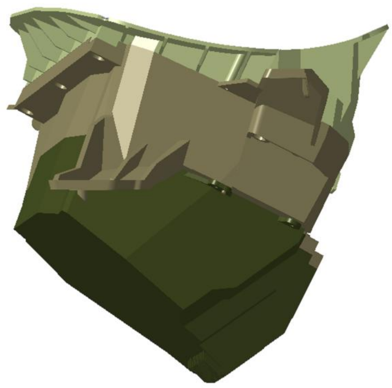


Front View



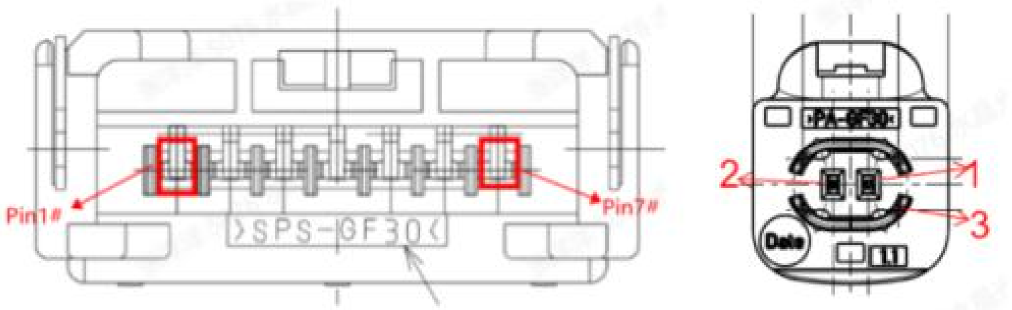
Back View

Left-Side



Right-Side

d) How many separate Connectors?	2



C1

C2

Connector Diagram

C1-Pin #	Signal Name	Wiring Detail (SW/TWP/STWP[1])
1	Battery Feed	SW
2	N/C	N/C
3	N/C	N/C
4	Input-Digital	SW
5	N/C	N/C
6	N/C	N/C
7	Ground	SW
C2-Pin #	**Signal Name**	**Wiring Detail (SW/TWP/STWP[1])**
1	FPD-IV+	STWP
2	FPD-IV-	STWP
3	Shield GND	STWP

[1]Abbreviations:
SW = Single Wire
TWP = Twisted Wire Pair
STWP = Shielded Twisted Wire Pair

1.4 Vehicle Packaging

a) Indicate the general package location of this Product in the Vehicle:									
Engine bay ▶	☐	Under body ▶	☐	Cabin ▶	☑	Boot ▶	☐	External Body ▶	☐

Provide further description below as necessary:
The HUD is mounted under the IP platform facing the driver, and only the Cover Lens and Top Housing are accessible in the cockpit.

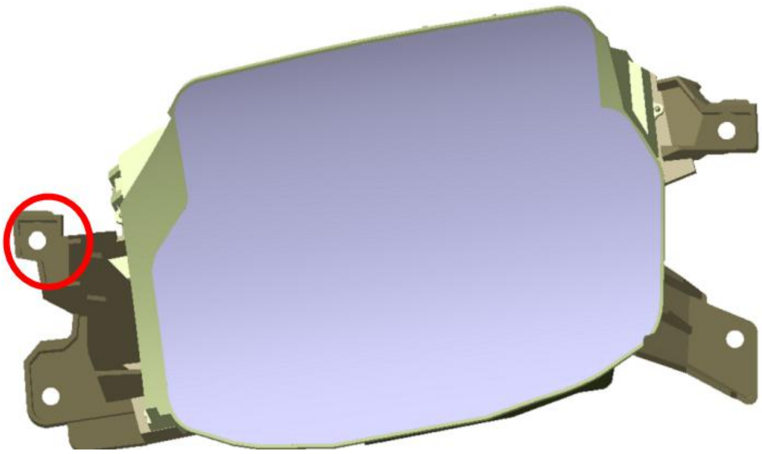
b) Will this Product be PACKAGED or TRANSPORTED in materials of differing Triboelectric Series values (either at tier 1 or tier 2 suppliers)?					
		Yes ▶	☐	No ▶	☑

c) What is the potential CUSTOMER ACCESSIBILITY to the Product:					
Access from Outside Vehicle Cabin ▶	☐	Access from Inside Vehicle Cabin ▶	☑	No Access ▶	☐
d) What is the nature of that Access:					
Direct Access (Physical touching) ▶	☑	Indirect Access (Remote Switch) ▶	☐		

Provide further description below as necessary:
Only the cover Lens of the HUD in the cabin can be touched

e) How is the Product connected to Power Return (GND)?				
Reference (GND):	Direct to Local Chassis▶	☐	In Harness (remote GND/Module) ▶	☑
Which Pin(s)?	C1 connector， Pin 7#			

f) How is the Case connected to a Reference Level (GND)?						
In the <u>Vehicle</u> – How is the metallic case connected?	Connect (GND) ▶	☑	Isolated ▶	☐	Unknown ▶	☐
In the <u>Product</u> – Is there a power return connection to the Case?	Direct (DC) ▶	☑	Coupled (AC) ▶	☐	NO Connect ▶	☐



Only the leg marked in the figure is grounded during the test.

g) Is the Product connected internally / externally to Magnetically Sensitive / Controlled Devices?							
Yes▶	☑	No▶	☐				
Internal ▶	☑	External ▶	☐	Sensitive ▶	☐	Controlled ▶	☐

Provide further description below as necessary:
The motor current is 55mA and the power is 0.67W

h) How is the Product connected to Vehicle Power?				
Please select which Vehicle power source(s) the product is connected to?		*List ALL Connector Pins that have an external connection to Vehicle Power*	Fused?	*Describe Fuse Rating for each connection*
Direct Battery Connection ▶	☑	*Pin 1 and pin 7 of connector C2*	☑	*10A*
‡ Switched Power 1 ▶	☐		☐	
‡ Switched Power 2 ▶	☐		☐	
‡ Switched Power 3 ▶	☐		☐	
‡ Switched Power 4 ▶	☐		☐	
Regulated Power ▶	☐			
† Internal Power ▶	☐	† Battery source internal to the DUT		

‡ Switched Power – any circuit connected to vehicle battery through a mechanical switch, electro-mechanical relay or electronic switch. Annotate the proper name for each Switched Power signal source (e.g. Run, Run/Start, Ignition, VPWR…).

2.0 EMC Requirements Analysis

2.1 Critical Interface Signals

Identify those signals whose EMC immunity is critical (potentially more susceptible, e.g. CAN, Vehicle Speed). For those critical signals, include electrical characteristics (e.g. Voltage/Current Level, Frequency, Duty Cycle).

Signal Description	Voltage/Current Level	Frequency	% Duty Cycle (range)	Other
FPD-LINK FPD-IV+	1.32V	192.8M	-	Signal
FPD-LINK FPD-IV-	1.32V	192.8M	-	Signal
Forward rate:6.75Gbps ; Backward rate:168.75Mbps				
Battery Feed	13.5V/2A	-	-	Power
Ground	0V	-	-	Ground
Input-Digital	12V/0.1A	-	-	Enable-Signal

2.2 Potential Sources of Emissions

List all DUT internal microprocessor clocks, subclocks, local oscillators etc. in addition to all periodic interface signals that may act as potential sources of radiated or conducted emissions e.g. PWM outputs. Signal characteristics including frequency, duty cycle, and signal voltage/current level should also be included.

Signal Source Description	Voltage/Current Level	Frequency	% Duty Cycle (range)	Other
Buck chip CLK	13.5V	850kHz	-	MCU and LCD Power supply
Buck chip CLK	3.3V	2640kHz	-	Deserializer Power supply
LED driver PWM	3.3V	2.44KHz	-	PWM signal of LED light driver chip
Step motor PWM	3.3V	1.2KHz	50	CLK signal of stepper motor
Crystal oscillator	1.4V	16MHz	50	MCU Clock
Crystal oscillator	1.8V	27MHz	50	Deserializer Clock
Crystal oscillator	1.6V	16MHz	50	Warping CLK
LCD CLK	1.25V	26.2MHz	50	LCD CLK
FPD-LINK FPD-IV±	1.32V	192.8MHz	-	FPD-LINK PCLK
I2C	3.3V	1MHz	-	Communication between Deserializer and MCU
SPI	3.3V	400KHz	-	Communication between Motor driver and MCU

2.3 Test Sample / Surrogate Selection

<table>
<tr><th>Product Part Number</th><th>Sample ID</th><th>For the test item</th><th>Remark</th></tr>
<tr><td rowspan="3">T2L2-14F906-BA</td><td>1#</td><td>All</td><td rowspan="3"></td></tr>
<tr><td>2#</td><td>All</td></tr>
<tr><td>3#</td><td>ESD</td></tr>
<tr><td rowspan="2">T2L2-14F906-AA</td><td>1#</td><td>ESD</td><td rowspan="2">In addition to structural parts mirror, other completely follow T2L2-14F906-BA</td></tr>
<tr><td>2#</td><td>ESD</td></tr>
<tr><td>T8U2-14F906-BA</td><td colspan="3">Equate to T2L2-14F906-BA</td></tr>
<tr><td>T8U2-14F906-AA</td><td colspan="3">Equate to T2L2-14F906-BA</td></tr>
</table>

3.0 Test Design and Requirements

3.1 DUT Operating Modes/Functional Classifications

List all DUT operating mode(s) that are active in each of the vehicle operating states (i.e. OFF, ACC, START, RUN). Place an "X" in the appropriate column to indicate the applicable vehicle operating states. For each DUT operating mode, list all major functions and place an "X" under the appropriate functional importance class. Use the table below for presentation of this information. All mode and function names listed in the table must include a subsequent description.

DUT Mode	Functions Active	Importance classification			Vehicle Operating Modes			
		Class A	Class B	Class C	Off	Accessory	Start	Run
Mode0	UN-Powered			×	×			
Mode1	Sleep			×		×		×
Mode2	Normal Operation and no motor operation			×		×		×
Mode3	Normal mode and continuous motor operation			×		×		×

Mode and Function Description(s):

DUT Mode	Mode 0: OFF/Unpowered	
Status	**Monitored by**	**Definition Description**
N/A	N/A	Function check after every handling electrostatic discharge test step, validate follow the mode 2.
DUT Mode	**Mode 1:** In sleep mode with no function (Power supply line connect to battery,MCU without power supply, Disconnect the enable line)	
Status	**Monitored by**	**Definition Description**
1.The Signal Sources BOX is powered by a separate battery. 2.Signal Sources BOX disconnect the enable line. 3.The monitoring camera is aim at the TFT screen of the HUD.	1.Monitoring camera 2.Sleep current	1. Observing the display screen of the surveillance camera through the human eye, the HUD cannot be awakened during the test. 2. Sleep current<100μA during the test.

<table>
<tr><td>DUT Mode</td><td colspan="2">Mode 2：
Normal Operation and no motor operation.
Power supply line connect to battery,enable line connect the signal sources BOX.
The HUD is awakened by the enable line and is in normal working state. The TFT screen can display driving, navigation and ADAS information.</td></tr>
<tr><th>Status</th><th>Monitored by</th><th>Definition Description</th></tr>
<tr><td>1.The Signal Sources BOX is powered by a separate battery.
2.Signal Sources BOX connect the enable line.
3.The monitoring camera is aim at the TFT screen of the HUD.</td><td>1.Monitoring camera
2.PC(upper computer software: The monitoring is as follows: MCU/ signal transmission/ memory and motor)</td><td>1. The HUD is set to nominal luminance, and then the monitoring camera is aim at the TFT screen of the HUD to observe the image quality in the display and evaluate it in a visually subjective manner:
Backlight：Constant level of backlight illumination during the test.
Display: If the image, noise, distortion, scintillation, freeze-out, and color variations are observed, then the quality of the image is considered to have been disturbed by the test conditions and it is deemed to be substandard.
2. Through the upper computer software on the PC side, the following functions are monitored during the test process：
MCU: According to the MCU heartbeat signal, it is determined whether the DUT has abnormal power-down during the test.
Signal Transmission: Determine whether the signal transmission is abnormal by monitoring the video lock.
Memory: During the test, the position of the motor is monitored to confirm whether the motor returns to the memory position after the interference disappears.
Motor：Monitor the motor position during the test. During the test, the motor does not work and it cannot be triggered by mistake.</td></tr>
<tr><td>DUT Mode</td><td colspan="2">Mode 3：
Normal Operation and continuous motor operation.
Power supply line connect to battery,enable line connect the signal sources BOX.
The HUD is awakened by the enable line and is in normal working state. The TFT screen can display driving, navigation and ADAS information.</td></tr>
<tr><th>Status</th><th>Monitored by</th><th>Definition Description</th></tr>
<tr><td>1.The Signal Sources BOX is powered by a separate battery.
2.Signal Sources BOX connect the enable line.</td><td>PC(upper computer software: The monitoring is as follows: MCU/ signal transmission/ memory and motor)</td><td>1. The HUD is set to nominal luminance.
2. Through the upper computer software on the PC side, the following functions are monitored during the test process：
PWM/ Backlight:By monitoring the real-time data of the PWM to determine whether there is a change from the input data before the test.
MCU: According to the MCU heartbeat signal, it is determined whether the DUT has abnormal power-down during the test.
Signal Transmission/ Display: Determine whether the signal transmission is abnormal by monitoring the video lock.
Memory: During the test, the position of the motor is monitored to confirm whether the motor returns to the memory position after the interference disappears.
Motor：During the test, the motor will move back and forth between the limits, monitor the dynamic motor position displayed in the upper computer, and confirm whether the motor is working normally.</td></tr>
</table>

3.2 Test Requirements

The DUT component / sub-system category is assessed as being:

Passive				Active					
P	R	BM	EM	A	AS	AM	AX	AY	AW
☐	☐	☑	☐	☐	☐	☐	☑	☐	☐

It is possible for multiple categories to be applicable.

Test Requirements – as defined in JLR-EMC-CS v1.0 unless otherwise indicated

Test Description	Test applies (Y/N)	Functional Class & Functional Status			Interface to be tested	S/C[1]	Operating Mode(s) to be used for indicated test
		A	B	C			
Radiated Immunity – RF							
RI 112 [4] Level 2 BCI	Y	II	II	I	All harnesses	S/C	Mode 1 \ 2 \ 3
RI 112 [4] Level 1 BCI	Y	I	I	I	All harnesses	S/C	Mode 1 \ 2 \ 3
RI 114 [4] Level 2 ALSE/Reverb	Y	II	II	I	DUT and all harnesses	C	Mode 1 \ 2 \ 3
RI 114 [4] Level 1 ALSE/Reverb	Y	I	I	I	DUT and all harnesses	C	Mode 1 \ 2 \ 3
RI 115 [4] Level 2 Portable Transmitter	Y	II	II	I	All DUT surfaces and harnesses	C	Mode 1 \ 2 \ 3
RI 115 [4] Level 1 Portable Transmitter	Y	I	I	I	All DUT surfaces and harnesses	C	Mode 1 \ 2 \ 3
RI 140 [4] Magnetic Field	N	I	I	I	All DUT surfaces	C	N/A

Note: Initially testing shall be performed using Level 2 requirements.

Test Description	Test applies (Y/N)	A	B	C	Interface to be tested	S/C	Operating Mode(s)
Coupled Immunity – RF (Not applicable to HV wires [3])							
RI 130 Inductive Transients	Y	I	I	I	All circuits	S	Mode 1 \ 2 \ 3
RI 150 Charging System	Y	I	I	I	All circuits	S	Mode 1 \ 2 \ 3
Conducted Immunity – Continuous (Not applicable to HV wires [3])							
CI 210 Continuous Disturbance	Y	I	I	I	*Power supply inputs* [2]	C	Mode 1 \ 2 \ 3

Test Description		Test applies (Y/N)	Functional Class & Functional Status			Interface to be tested	S/C[1]	Operating Mode(s) to be used for indicated test
			A	B	C			
Conducted Immunity – Transients (Not applicable to HV wires [3])								
CI 220 Pulse A1	Switched Power < 5A	N	II	II	II	*Power supply inputs [2]*	S	N/A
	Control Circuits	N	II	II	II	*Power supply inputs [2]*	S	N/A
CI 220 Pulse A2-1	Switched Power < 5A	N	II	II	II	*Power supply inputs [2]*	S	N/A
CI 220 Pulse A2-1 Pulse A2-2	Control Circuits	N	II	II	II	*Power supply inputs [2]*	S	N/A
CI 220 Pulse C-1		Y	I	I	I	*Power supply inputs [2]*	S	Mode 2 and 3
CI 220 Pulse C-2		Y	I	I	I	*Power supply inputs [2]*	S	Mode 2 and 3
CI 220 Pulse E	Switched Power ≥ 5A	N	II	II	II	*Power supply inputs [2]*	S	N/A
	Control Circuits	N	II	II	II	*Power supply inputs [2]*	S	N/A
CI 220 Pulse F1		Y	I	I	I	*Power supply inputs [2]*	S	Mode 2 and 3
CI 220 Pulse F2		Y	II	II	II	*Power supply inputs [2]*	S	Mode 2 and 3
CI 220 Pulse G1 (Normal Load Dump)		N	III	III	II	*Power supply inputs [2]*	C	N/A
CI 220 Pulse G2 (Central Load Dump)		Y	III	III	II	*Power supply inputs [2]*	C	Mode 2 and 3
Conducted Immunity - Power Cycle (Not applicable to HV wires [3])								
CI 230 Waveform A		N	II	II	II	*Power supply inputs not active during start [2]*	C	N/A
CI 230 Waveform B		N	II	II	II	*Ignition power supply inputs [2]*	C	N/A
CI 230 Waveform C		N	II	II	II	*Power supply inputs only active during start [2]*	C	N/A
CI 230 Waveform D		Y	II	II	II	*Direct battery power supply inputs [2]*	C	Mode 2 and 3
Conducted Immunity - Voltage Offset (Not applicable to HV wires [3])								
CI 250 (setup a)		Y	I	I	I	*All DUT ground inputs [2]*	S	Mode 2 and 3
CI 250 (setup b)		N	I	I	I	*All external load/sensor grounds [2]*	S	N/A

Test Description		Test applies (Y/N)	Functional Class & Functional Status			Interface to be tested	S/C[1]	Operating Mode(s) to be used for indicated test
			A	B	C			
Conducted Immunity - Voltage Dropout (Not applicable to HV wires [3])								
CI 265 Waveform A		Y	II	II	II	*Power supply inputs [2]*	C	Mode 2 and 3
CI 265 Waveform B		Y	II	II	II	*Power supply inputs [2]*	C	Mode 2 and 3
CI 265 Waveform C		Y	II	II	II	*Power supply inputs [2]*	C	Mode 2 and 3
CI 265 Waveform D		Y	I	I	I	*Power supply inputs [2]*	S	Mode 2 and 3
Conducted Immunity - Voltage Overstress (Not applicable to HV wires [3])								
CI 270- A **-14 Volt** Reverse Battery		Y	III	III	III	*Power supply inputs [2]*	C	Mode 2 and 3
CI 270- B **+19 Volt** Over voltage – Failed Regulator		Y	III	II	II	*Power supply inputs [2]*	C	Mode 2 and 3
CI 270- C **+ 28 volt** Over voltage – Jump Start	BATT / IGN ≥ 60 secs	Y	III	I/II	I/II	*Power supply inputs [2]*	C	Mode 2 and 3
	START ≥ 15 secs	N	III	I/II	I/II	*Power supply inputs [2]*	C	N/A
Conducted Emissions - Transient (Not applicable to HV wires [3])								
CE 410 – Transient		Y	I	I	I	*Power supply inputs [2]*	S	Mode 3
Conducted Emissions - RF								
CE 420 [4] - RF Seq. CP01 to CP02		Y	I	I	I	Each power supply inputs and each harness	S/C	Mode 3
Conducted Immunity – Electrostatic Discharge								
CI 280 Handling (DUT not powered) Seq.U-01 to U-04		Y	IV	IV	IV	*All DUT surfaces and connector pins*	S	Mode 0
CI 280 Powered Seq. DP-01 to DP-04		Y	I	I	I	*All DUT surfaces and connectors*	S	Mode 2 and 3
CI 280 Powered Sequence DP-05 (15 KV)		Y	II	II	II	*All DUT surfaces and connectors accessible from vehicle interior*	S	Mode 2 and 3
CI 280 Powered Sequence DP-06 (25 KV)		N	II	II	II	*All DUT surfaces and connectors accessible from vehicle exterior*	S	N/A
CI 280 Powered Seq. IP-01 to IP-03		Y	I	I	I	*Discharge Islands*	C	Mode 2 and 3

Test Description	Test applies (Y/N)	Functional Class & Functional Status			Interface to be tested	S/C[1]	Operating Mode(s) to be used for indicated test
		A	B	C			
Radiated Emissions							
RE 310 [4] (0.15 - 5905 MHz)	Y	I	I	I	*DUT and all circuits*	C	Mode 3
RE 320 [4] (20 Hz – 150 kHz)	Y	I	I	I	*DUT and all circuits*	C	Mode 3

† - Test levels to be specified by the JLR HV Subsystems team.

NOTES

1. Indicate specific DUT circuit interface(s) to be subjected to test.
"C" (Combined): Indicates circuits are to be tested simultaneously.
"S" (Separate): Indicates circuits are to be tested separately.

2. List specific power supply and/or ground inputs to be subjected to testing (including the connector pin number). Where a test is not applicable (i.e. Test applies = N) .

3. HV supply lines are defined as any voltage supply with amplitude of greater than 60V dc and lower than 1500V dc (e.g. electric/hybrid vehicle traction battery supply)

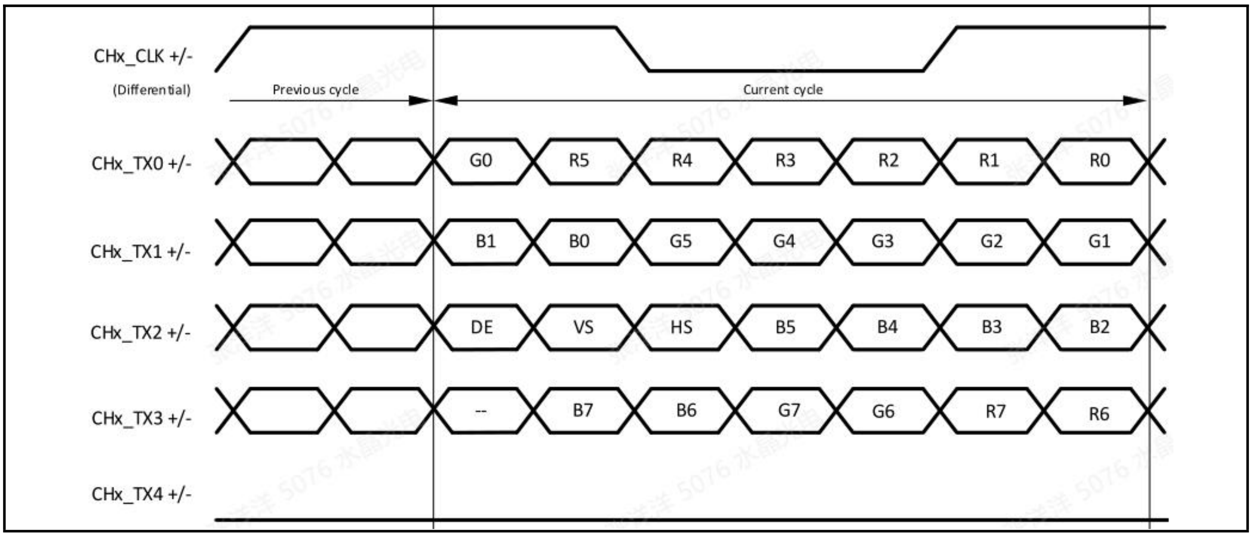
4. The HV component test setups for tests marked with this note shall be in line with CISPR 25 Ed 4 Annex I

3.3 Input Requirements

Select input conditions that will place the DUT in the desired operating mode(s) required for each test listed in section 3.2. List all modes required and the signal names used in them. Duplicate entries for signal names may exist but under different modes. Include any additional information that is needed to support operation of the DUT during testing including data bus messages, special test software, and/or any non-electrical interfaces. The format below should be used to provide this information.

Electrical Input Signals/Characteristics to Operate DUT in the specified test Mode

DUT Mode	Signal Name	Test	Pin #	Waveform	Amplitude	Freq/PW/DC%	Other
ALL	Battery Feed(C1)	ALL TEST	1	DC	13.5v	-	-
Mode1	Input-Digital (C1)	ALL TEST	4	-	-	-	-
Mode2	Input-Digital (C1)	ALL TEST	4	-	13.5v	-	-
ALL	Ground(C1)	ALL TEST	7	DC	Ground	-	-
ALL	FPD-IV+(C2)	ALL TEST	1	-	1.32v	-	-
ALL	FPD-IV-(C2)	ALL TEST	2	-	1.32v	-	-
Waveform of FPD:							



Non-electrical input signals/characteristics to make DUT functional:

- The signal source box transmits the image signal through FPD link；
- The signal source box controls the enable line of the HUD through Input-Digital；
- The PC sends instructions to the image source box through the upper computer software, and the image source box can control the work of the motor in the non-boot and shutdown states through FPD link；

3.4 Output Requirements

Select output signals or non-electrical indicators that will be monitored to verify the required functionality of the DUT for each operating Mode(s) selected in section 3.2. List signals or non-electrical indicators using the table format below. The acceptance criteria chosen shall equal the upper limit of deviation where the degradation in functionality becomes perceivable to the customer. The nominal and acceptable deviation values provided here define Function Performance Status I for this product.

Electrical output(s) to monitor and acceptance criteria

Motor operating mode	Current monitoring	Notes
No motor operation	When the motor is in an inoperative state, the motor should not be triggered by mistake during the test, and the HUD current value should not change. If the motor is triggered by mistake, the current will increase(approximately 30mA to 63mA, only used as a reference, and the final current is based on the actual operation of the motor).	
Continuous motor operation	When the motor is in a continuous working state, The current of the HUD will increase or decrease regularly(approximately 30mA to 63mA, only used as a reference, and the final current is based on the actual operation of the motor).	

CI 280 ESD Parametric Value Requirements

(add measurements for the lab to identify potential ESD damage)

Parameter Name	Nominal Value	Tolerance	Reason for change within tolerance band	Reason for change out of tolerance band	Notes
Normal operation current	<2.2A[Note2]	±10%	part to part variation	Major damage	Keep the test environment consistent before and after the test: voltage 13.5V, room temperature and the same brightness
Sleep current	<100uA	±10%	part to part variation	Major damage	Verify the value before and after the test
Battery Feed to GND	19MΩ	±10%	part to part variation	Major damage	Measure the LCR value of the PIN before and after the test
Input-Digital to GND	2.2nF	±25%	part to part variation	Major damage	
FPD-LINK FPD-IV+ to GND	100nF	±25%	part to part variation	Major damage	
FPD-LINK FPD-IV- to GND	100nF	±25%	part to part variation	Major damage	

Note：

1. Before verifying the values in the table above, the laboratory needs to perform functional inspections of Mode 1, Mode 2 and Mode 3 of the prototype respectively. Its functional performance and upper computer data must meet the requirements of Section 3.1.
2. The operational current of the HUD shall not exceed 2.2A,The laboratory should compare the initial and final measurements, the change should be within ±10% range, and the measurement should be less than 2.2A.

The additional parameter test list for teardown analysis：

Parameter Name	Nominal Value	Tolerance	Reason for change within tolerance band	Reason for change out of tolerance band	Notes
3.3V for MCU (C4012)	3.3V	±0.15V	allowable value before and after applying ESD	Major damage	Battery Voltage 13.5V
1.15V for Deserializer (C9498)	1.15V	±0.04V	allowable value before and after applying ESD	Major damage	Battery Voltage 13.5V
1.8V for Deserializer (C4201)	1.8V	±0.11V	allowable value before and after applying ESD	Major damage	Battery Voltage 13.5V
Motor Driver (C8513)	13.5V	±1.5V	allowable value before and after applying ESD	Major damage	Battery Voltage 13.5V
LVDS (L9350-L9354)	For detailed data, see Appendix A, chapter 2.1		allowable value before and after applying ESD	Major damage	
TVS (TVS8001 TVS8000 TVS5250 TVS5251)	For detailed data, see Appendix A, chapter 2.7		allowable value before and after applying ESD	Major damage	
LED Driver (C9425)	24V	±25%	allowable value before and after applying ESD	Major damage	Need to be tested at maximum brightness

For supplementary content refer to Appendix A

Non-electrical output(s) to monitor and acceptance criteria

(e.g. instrument cluster visual display or LED illumination intensity)

Mode	Function Description	Test	Note 2	Acceptance criterion for function
OP1 (sleep)	Backlight/ Display	All	N	ECU powered and in sleep mode
		RI/CI	A	Functional Performance Status I: No wake up allowed. Functional Performance Status II: Neither display nor backlight activation is allowed. DUT may wake up but it shall self-recover after test and enter sleep mode immediately after disturbance disappears in line with function's safe recovery strategy.
OP2 (Normal mode)	Backlight	All	N	Constant level of backlight illumination
			A	Functional Performance Status I :No distortion and no flickering allowed. The function shall operate as designed. Functional Performance Status II : Change of illumination level is allowed, but it shall be reviewed and approved by JLR. DUT shall self-recover to normal operation when disturbance is removed. Functional Performance Status III : Change/loss of illumination is allowed, but it shall be reviewed and approved by JLR. DUT shall recover to normal operation when disturbance is removed. Allow recovery from soft start (power restarts are not allowed).
	Display	All	N	Distortion/flicker free display content
			A	Functional Performance Status I :No noise/ distortion/ flickering/ freezing/ colour change is allowed. The function shall operate as designed. Functional Performance Status II : Distortion and degradation is allowed, but it shall be reviewed and approved by JLR. DUT shall self-recover to normal operation when disturbance is removed. Functional Performance Status III : Distortion, freeze, or loss of video image is allowed, but it shall be reviewed and approved by JLR. DUT shall recover to normal operation when disturbance is removed. Allow recovery from soft start (power restarts are not allowed).
	MCU	All	N	MCU heartbeat signal is sent and received with no errors.
			A	Status I: No error frames to be transmitted and all received data to be correctly interpreted. Status II: No error frames allowed. Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Status III: Some temporary interruption to data traffic or error frames may be allowed, but must recover to normal operation when disturbance is removed and/or following operator action. Any abnormal performance needs to be reviewed and approved by JLR.

Mode	Function Description	Test	Note 2	Acceptance criterion for function
OP2 (Normal mode)	Signal Transmission	All	N	Send and receive video and backchannel messages with no error frames.
			A	Status **I**: No error frames to be transmitted and all received data to be correctly interpreted. Status **II**: No error frames allowed. Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Status **III**: Some temporary interruption to data traffic or error frames may be allowed, but must recover to normal operation when disturbance is removed and/or following operator action. Any abnormal performance needs to be reviewed and approved by JLR.
	Motor	All	N	The motor will not be triggered by mistake during the test.
			A	Functional Performance Status **I** :It is not allowed to rotate when the motor is in a non-working state during the test. Functional Performance Status **II** :Feedback of motor position data is not allowed. Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Functional Performance Status **III** :Allows feedback of motor position data, but normal operation must be restored when the interference is removed and/or the operator's actions are followed.Any abnormal performance needs to be reviewed and approved by JLR.
	Memory	All	N	Memory function is not affected.
			A	Functional Performance Status **I** :The memory function shall operate as designed during and after exposure to a disturbance. Functional Performance Status **II** :It is not allowed for memory function to be affected.Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Functional Performance Status **III** :Allows some delays and deviations in memory function,but normal operation must be restored when the interference is removed and/or the operator's actions are followed.Any abnormal performance needs to be reviewed and approved by JLR.

Mode	Function Description	Test	Note 2	Acceptance criterion for function
OP3 (Normal mode+ Motor Operation)	MCU	All	N	MCU heartbeat signal is sent and received with no errors.
			A	Status I: No error frames to be transmitted and all received data to be correctly interpreted. Status II: No error frames allowed. Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Status III: Some temporary interruption to data traffic or error frames may be allowed, but must recover to normal operation when disturbance is removed and/or following operator action. Any abnormal performance needs to be reviewed and approved by JLR.
	Signal Transmission/ Display	All	N	Send and receive video and backchannel messages with no error frames.
			A	Status I: No error frames to be transmitted and all received data to be correctly interpreted. Status II: No error frames allowed. Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Status III: Some temporary interruption to data traffic or error frames may be allowed, but must recover to normal operation when disturbance is removed and/or following operator action. Any abnormal performance needs to be reviewed and approved by JLR.
	Motor	All	N	During the test, the motor rotates between the minimum gear and the maximum gear as expected, without error.
			A	Status I: No error reported during the test. Status II: No error is reported, but a short caton is allowed, and normal can be automatically restored after the interference disappears.. Any abnormal performance needs to be reviewed and approved by JLR. Status III: Allow errors and long caton,but must recover to normal operation when disturbance is removed and/or following operator action. Any abnormal performance needs to be reviewed and approved by JLR.
	Memory	All	N	Memory function is not affected.
			A	Functional Performance Status I :The memory function shall operate as designed during and after exposure to a disturbance. Functional Performance Status II :It is not allowed for memory function to be affected.Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed. Functional Performance Status III :Allows some delays and deviations in memory function,but normal operation must be restored when the interference is removed and/or the operator's actions are followed.Any abnormal performance needs to be reviewed and approved by JLR.
	Backlight	All	N	PWM real-time data will not change
			A	Functional Performance Status I :PWM real-time data will not deviate from the set data during and after the test. Functional Performance Status II :PWM data changes are not allowed.Any abnormal performance needs to be reviewed and approved by JLR. DUT must self-recover to normal operation when disturbance is removed.

				Functional Performance Status **III** :Delays and changes are allowed,but normal operation must be restored when the interference is removed and/or the operator's actions are followed.Any abnormal performance needs to be reviewed and approved by JLR.

*Example for guidance – please delete.

Note2:
- **N** = Nominal Value
- **A** = Acceptable Value

3.5 Load Box/Test Support Requirements

List all test fixture information. For each DUT circuit include the pin number, its name and description. Indicate by placing an "X" under the appropriate column whether it is an input or output and if loaded, is it connected to a real or simulated termination. Also include information on additional support hardware/software requirements. Include detailed block diagrams of the load box and/or support equipment. Specify how any support equipment used will be configured so as not to influence the test results.

Signal Sources BOX Description:
The signal source box is used as an auxiliary tooling to support HUD for EMC testing.The HUD itself cannot generate an image signal, so the signal source box needs to simulate the body CCCM to provide signal switches and image information for the HUD.

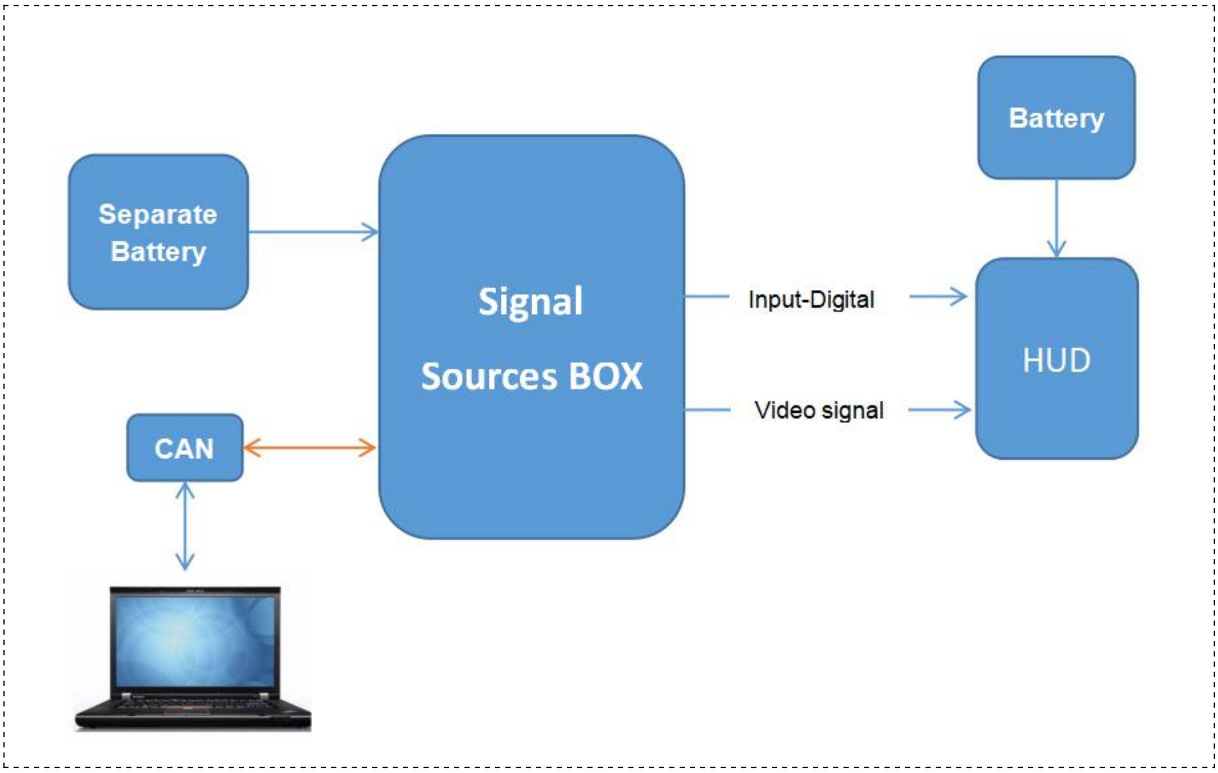
Signal Sources BOX connector 1:

Name	Description	Pin #	Load Value	Input	Output	Simulated	Actual	Reference
VBAT	Power supply	1	/	X	-	-	-	-
Enable_line	Enable	2	/	-	X	CCCM-EN	Enable	-
CAN_H	CAN	3	/	X	X	-	Test monitoring	-
CAN_L	CAN	4	/	X	X	-		-
GND	GND	5	/	-	-	-	-	-
NC	NC	6	/	-	-	-	-	-
NC	NC	7	/	-	-	-	-	-
NC	NC	8	/	-	-	-	-	-
GND	GND	9	/	-	-	-	-	-
NC	NC	10	/	-	-	-	-	-
NC	NC	11	/	-	-	-	-	-
GND	GND	12	/	-	-	-	-	-

Signal Sources BOX connector 2:

Name	Description	Pin #	Load Value	Input	Output	Simulated	Actual	Reference
FPD_DOUT+	Fpd-link signal out	1	-	-	X	Video/Data link	Video/ Data link	-
FPD_DOUT-	Fpd-link signal out	2	-	-	X	Video/Data link	Video/ Data link	-

Signal Sources BOX Diagram



4.0 DUT Test Set-up Diagrams

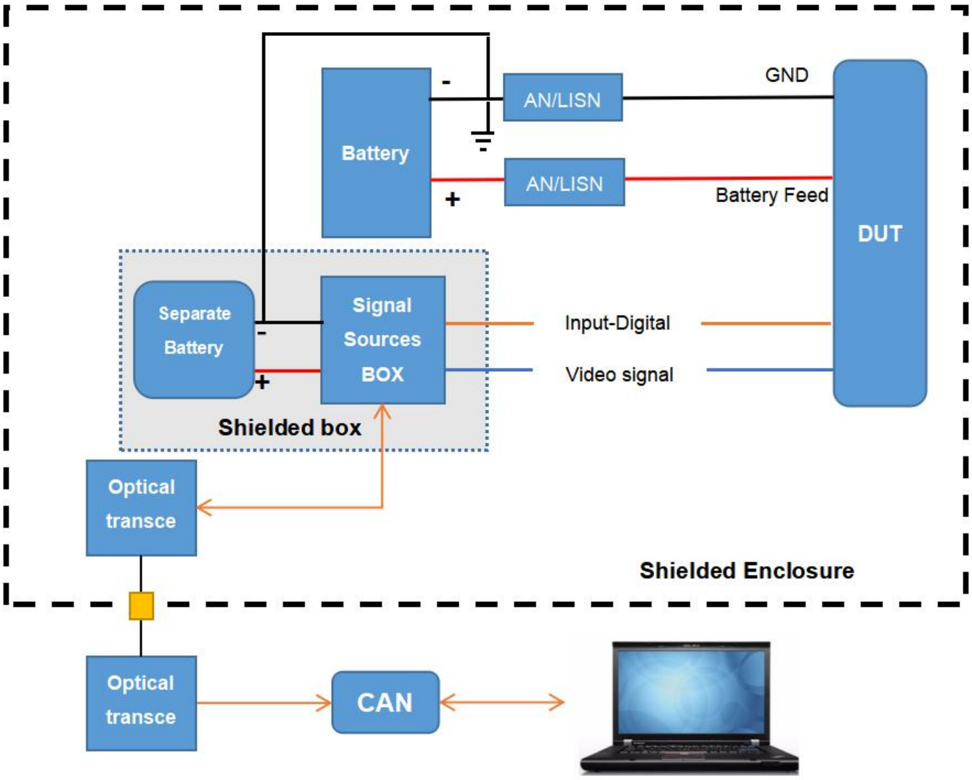
Detailed Test Setup – Informative

It is recognised that approved test facilities are capable of configuring the component/sub-system to be tested in accordance with the generic test set-ups illustrated in the EMC specification, JLR-EMC-CS. It is therefore not necessary in most cases to provide detailed test set-up information for all tests.

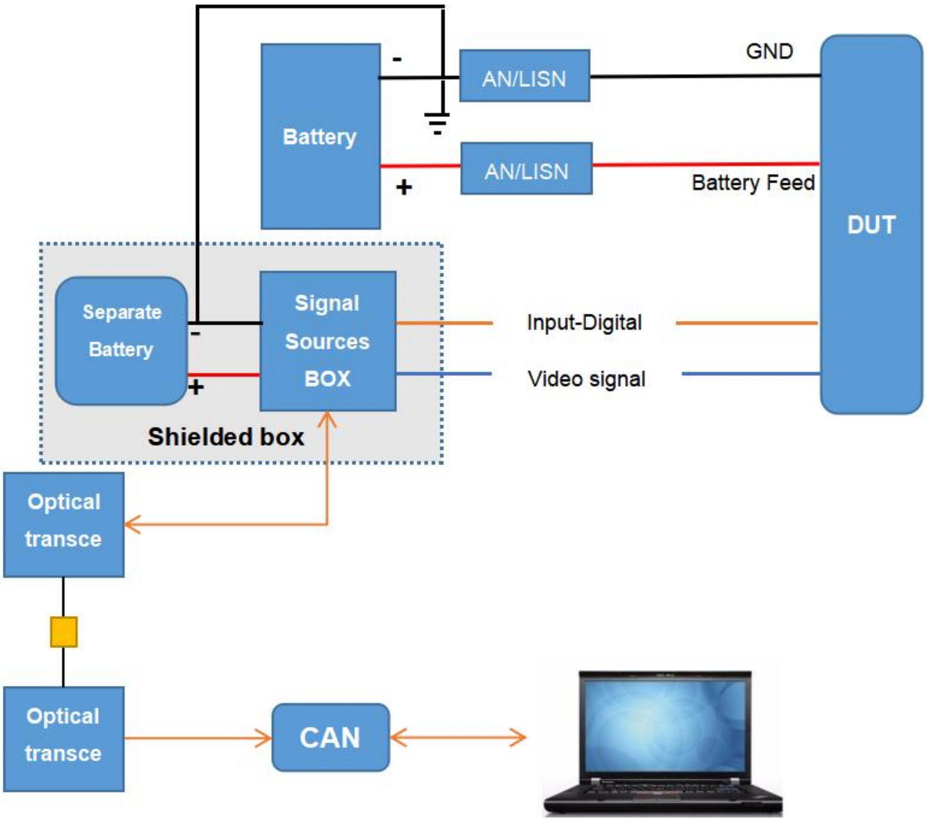
Occasionally however, specific test exceptions / differences from the generic test set-ups will be necessary e.g. the DUT is too large for mounting on the ground plane or requires specialised support equipment. In these cases detailed test setup information must be documented in this section. The table below shall be used to indicate where these exceptions apply.

Note: ESD test point illustrations shall always be included, even if there is no deviation from the specified test method.

Test ID	Requirement / Description	Deviation from specified test method. (Y/N)	Comments
CI 280 (U)	Electrostatic Discharge (Un-Powered / Handling)	N	*Must include to show ESD test points*
CI 280 (P)	Electrostatic Discharge (Powered)	N	*Must include to show ESD test points*
RI 112	RF Immunity - BCI	N	*FPD line only measure CBCI*
RI 114	RF Immunity – ALSE/Reverberation	N	*Without 600V/m* *At test frequencies 1000 MHz, the DUT shall be tested in a minimum of three (3) orthogonal orientations*
RI 115	RF Immunity – Hand Portable Transmitter	N	
RI 130	Coupled Immunity – Inductive Transients	N	
RI 140	Magnetic Field Immunity	/	
RI 150	Coupled Immunity – Charging	N	
CI 210	Conducted Immunity – Continuous Disturbances	N	
CI 220	Conducted Immunity – Transient Disturbances	N	
CI 230	Conducted Immunity – Power Cycling	N	
CI 250	Conducted Immunity – Voltage Offset	N	
CI 265	Conducted Immunity – Voltage Dropout	N	
CI 270	Conducted Immunity – Voltage Overstress	N	
RE 310	Radiated RF Emissions	N	
RE 320	Radiated RF Emissions	N	
CE 410	Conducted Transient Emissions	N	
CE 420	Conducted RF Emissions (LV)	N	



Test in Shielded Enclosure

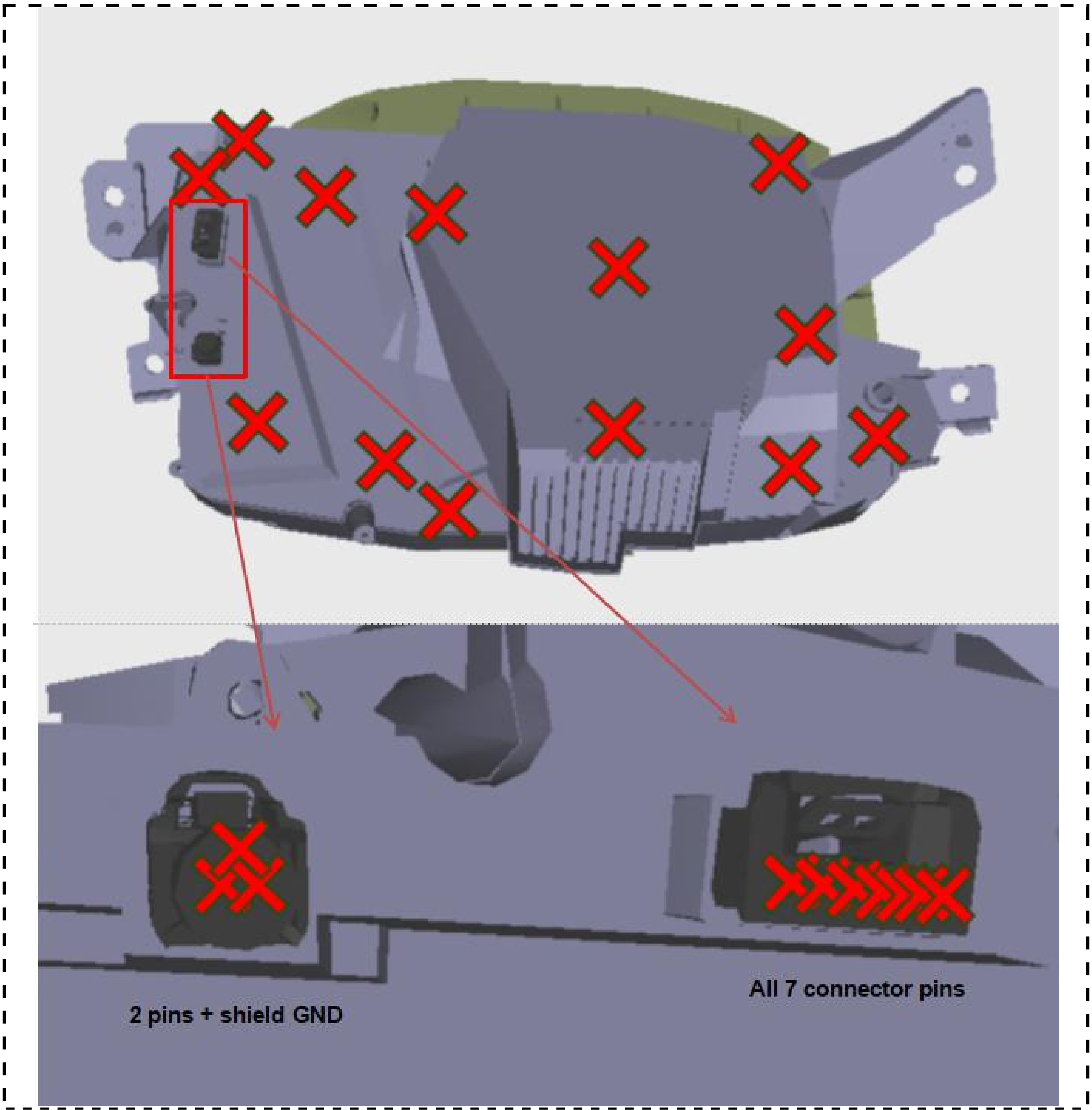


Test in Non-Shielded Enclosure

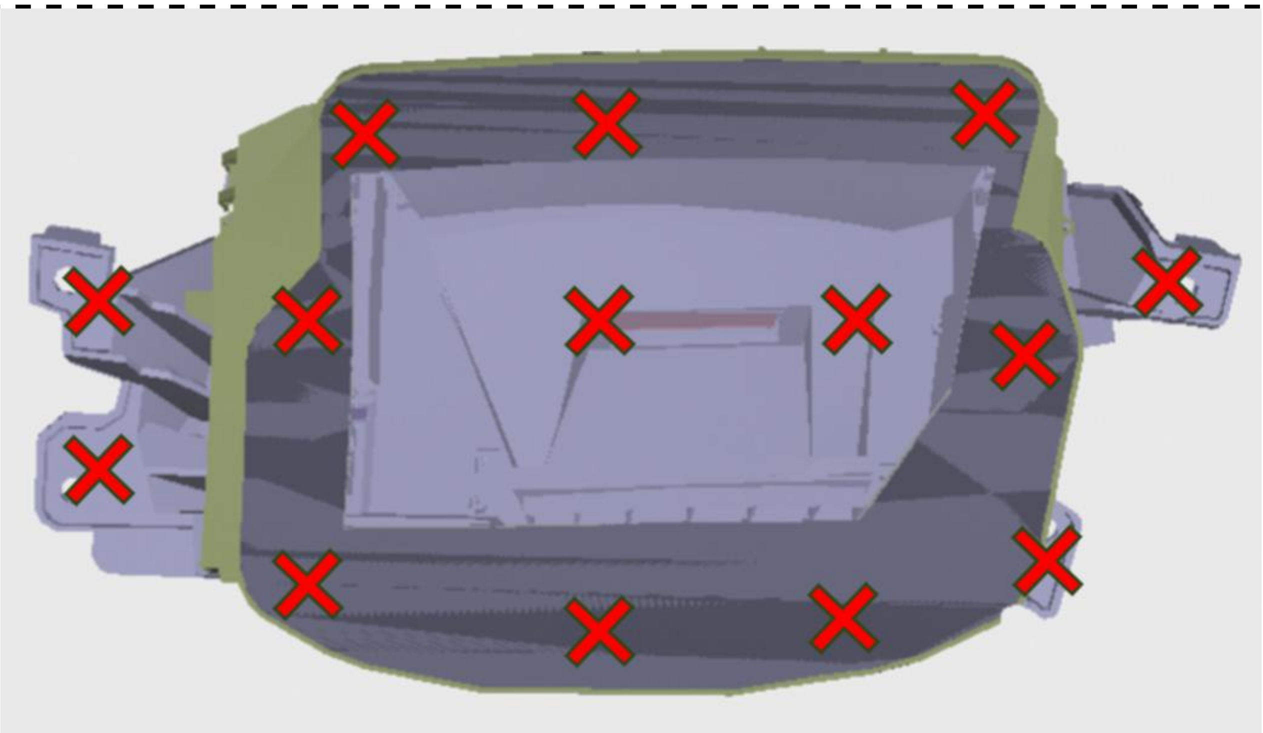
These two test set-up diagrams are the DUT test in the Shielded room and the DUT test in the Unshielded room.

Test Set-up Diagrams

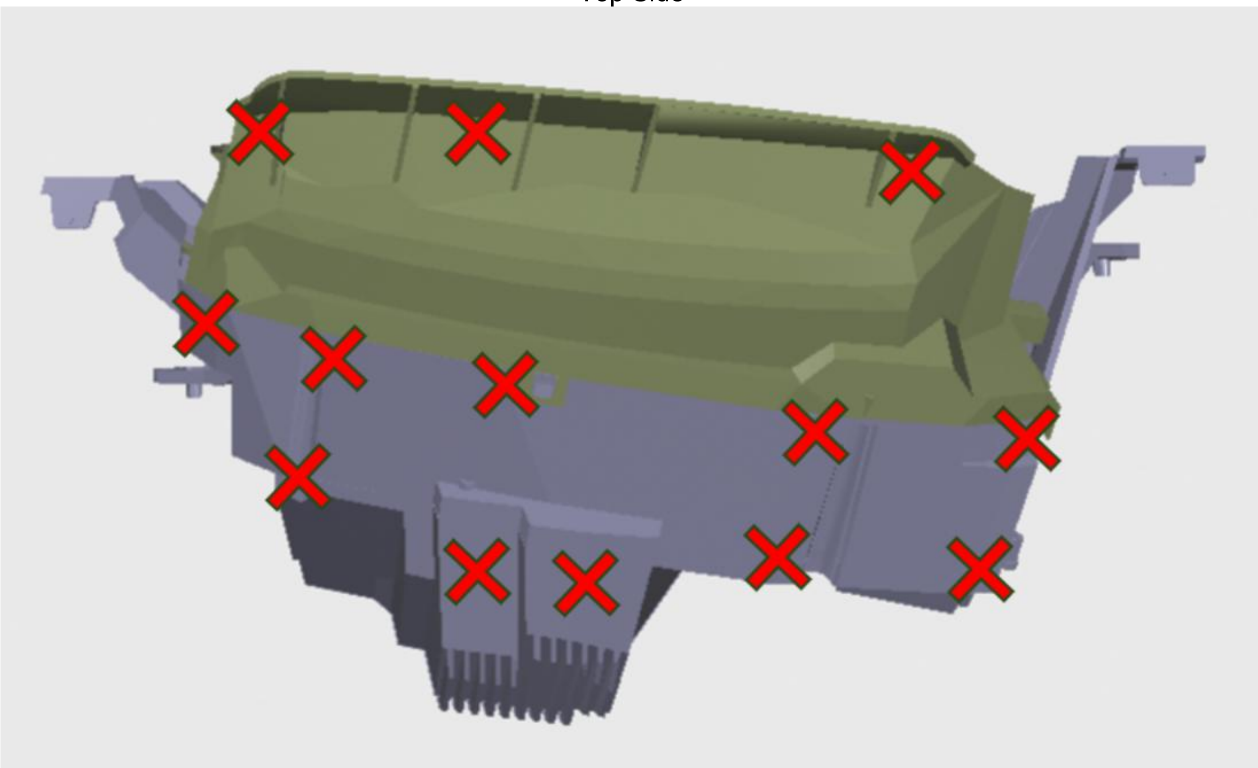
4.1 Electrostatic Discharge (CI 280:unpowered)



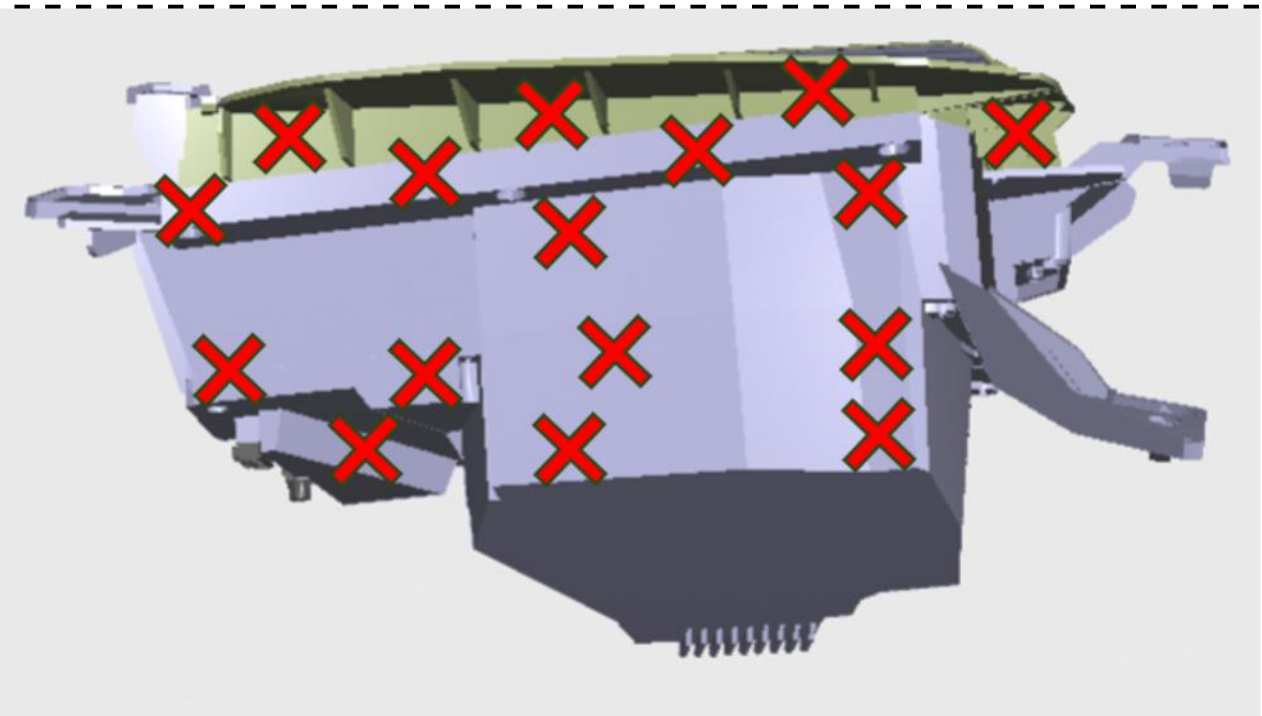
Bottom side and PIN



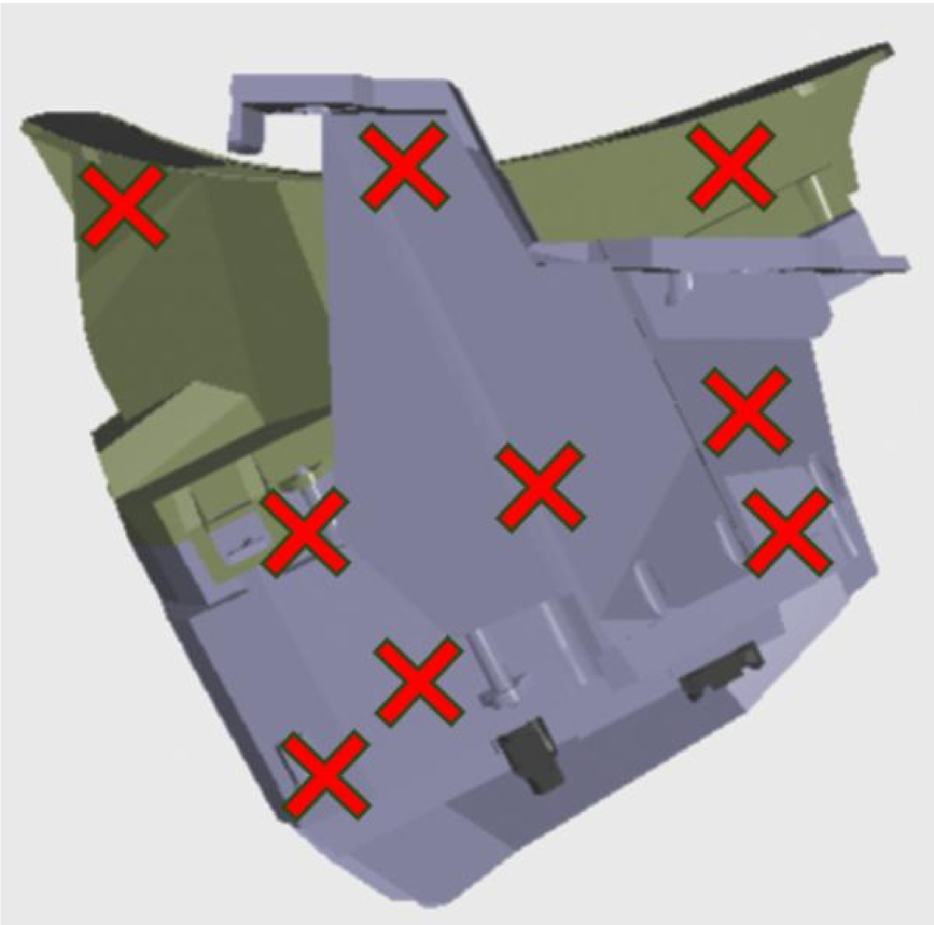
Top Side



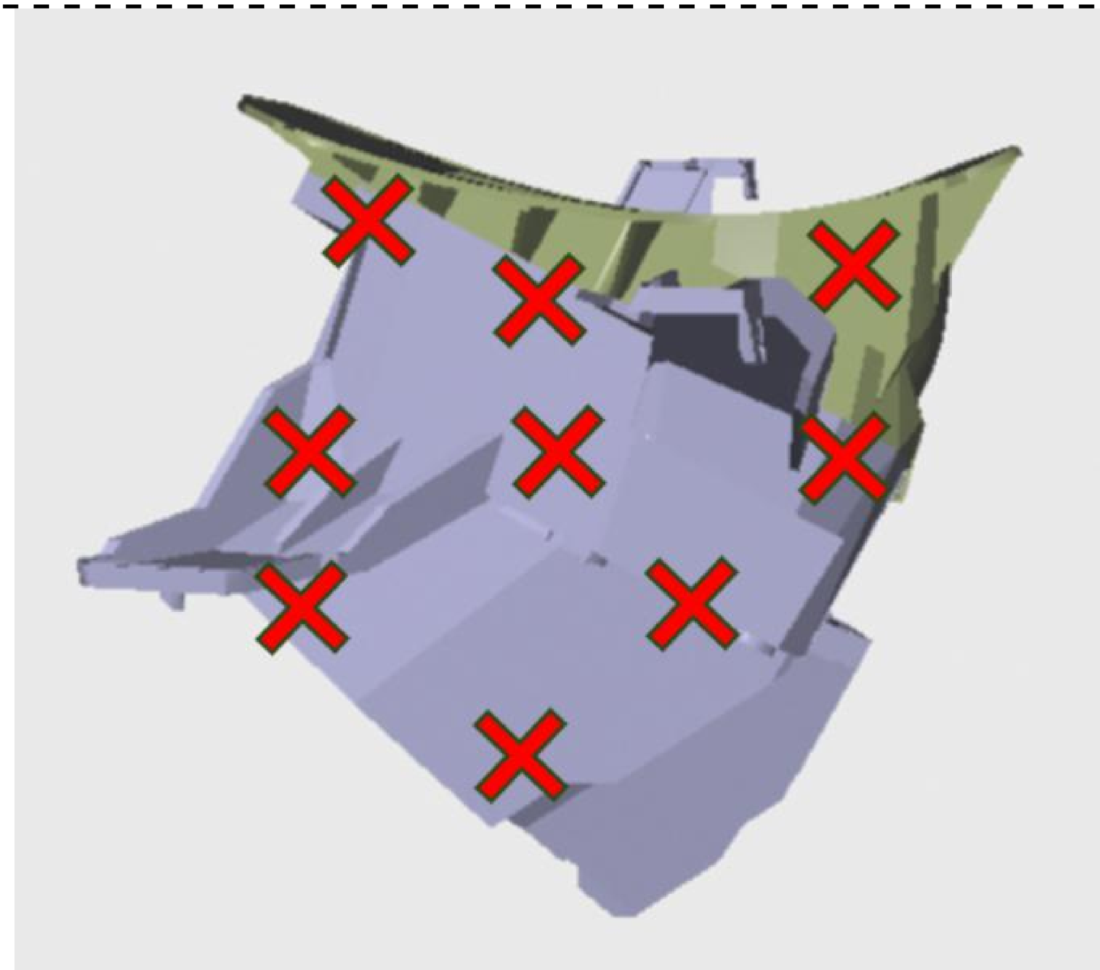
Front Side



Back Side



Left Side



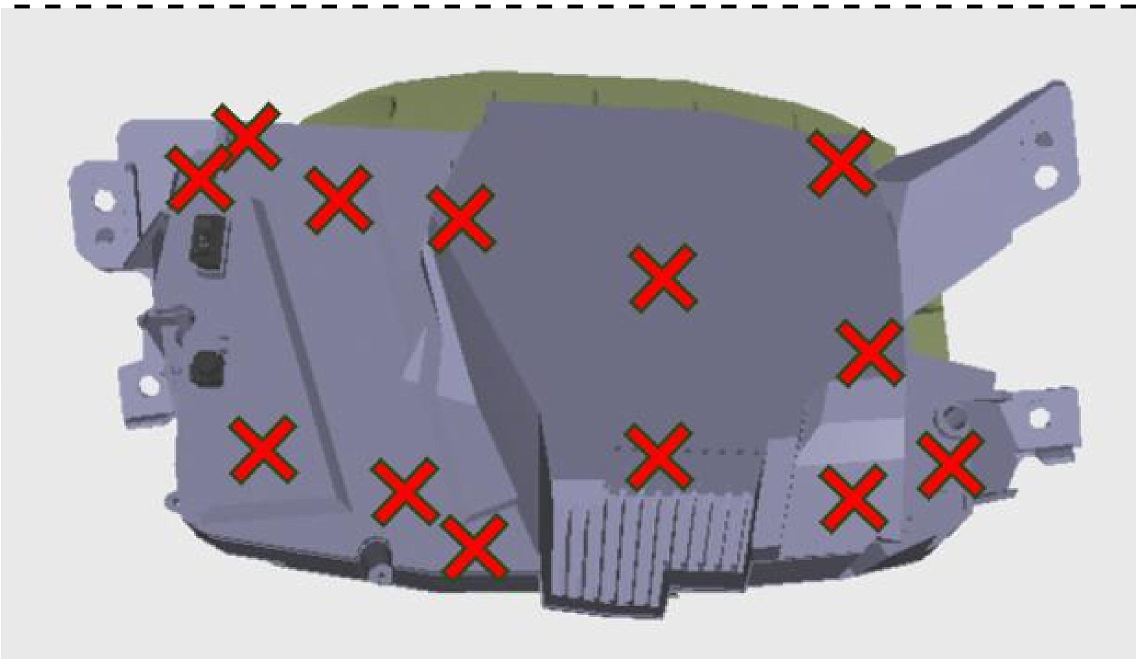
Right Side

Red points represent both air and contact discharges

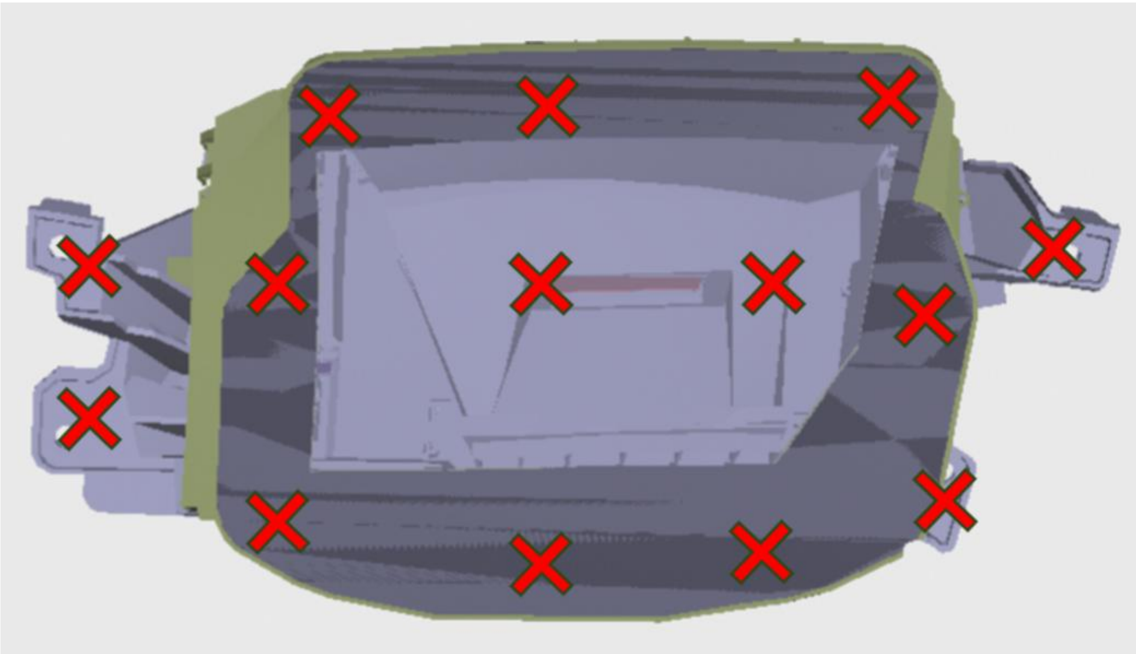
ESD test point

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 20.0 (UN-powered) ISO 10605
Deviations from Specified Test Method	N/A
Harness Configuration	No Harness
DUT Orientation	See the ESD test points
DUT Grounding (case or harness)	The case is not grounded. Instead the ground pin 7#（C1）
Additional Test Specific Information	When applying discharges to the DUT connector pins, All DUT power return terminals shall be connected to the ground plane via a grounding strap or wire with a maximum length of 200 mm
DUT Monitoring Information	The test process is not monitored, and verify the function after testing

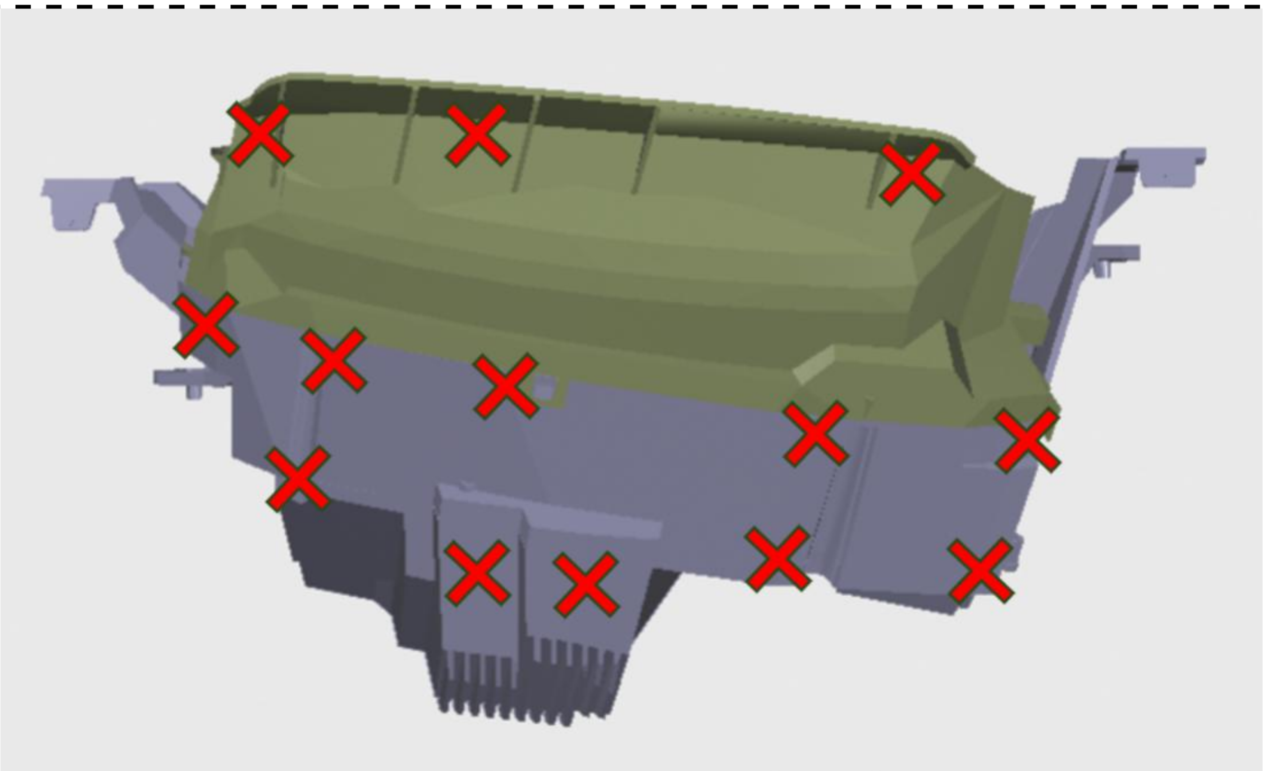
4.2 Electrostatic Discharge (CI 280: powered)

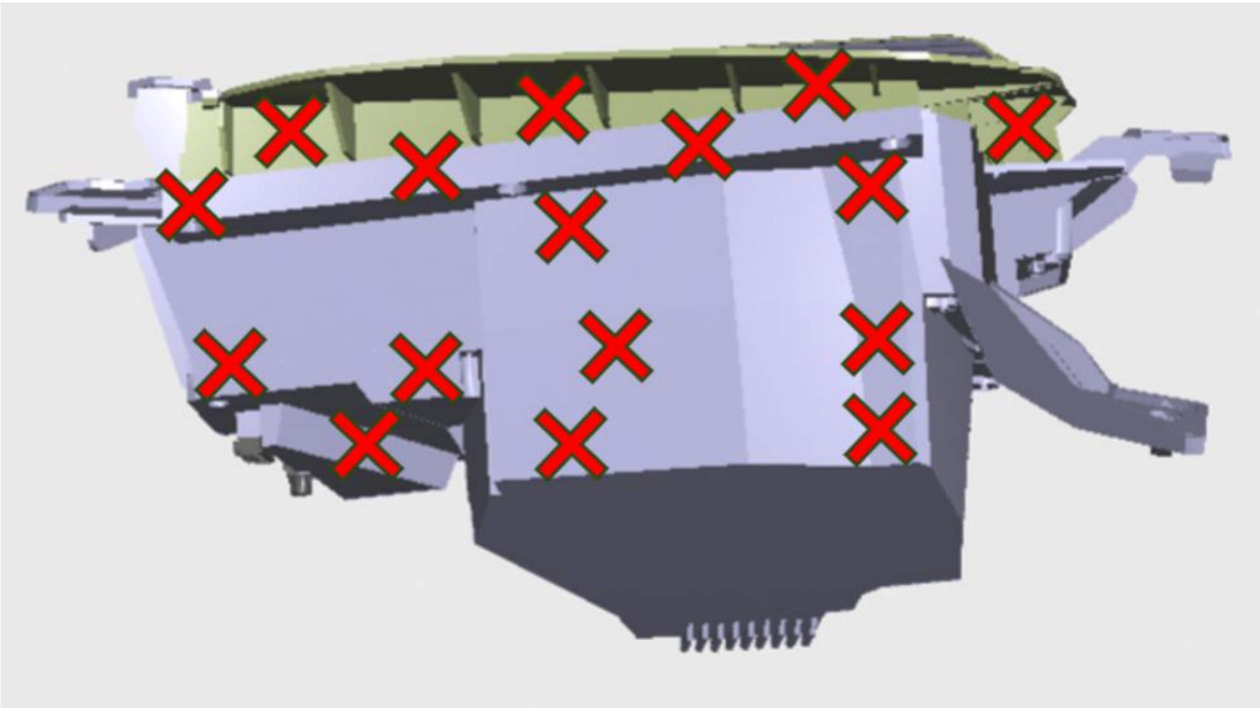


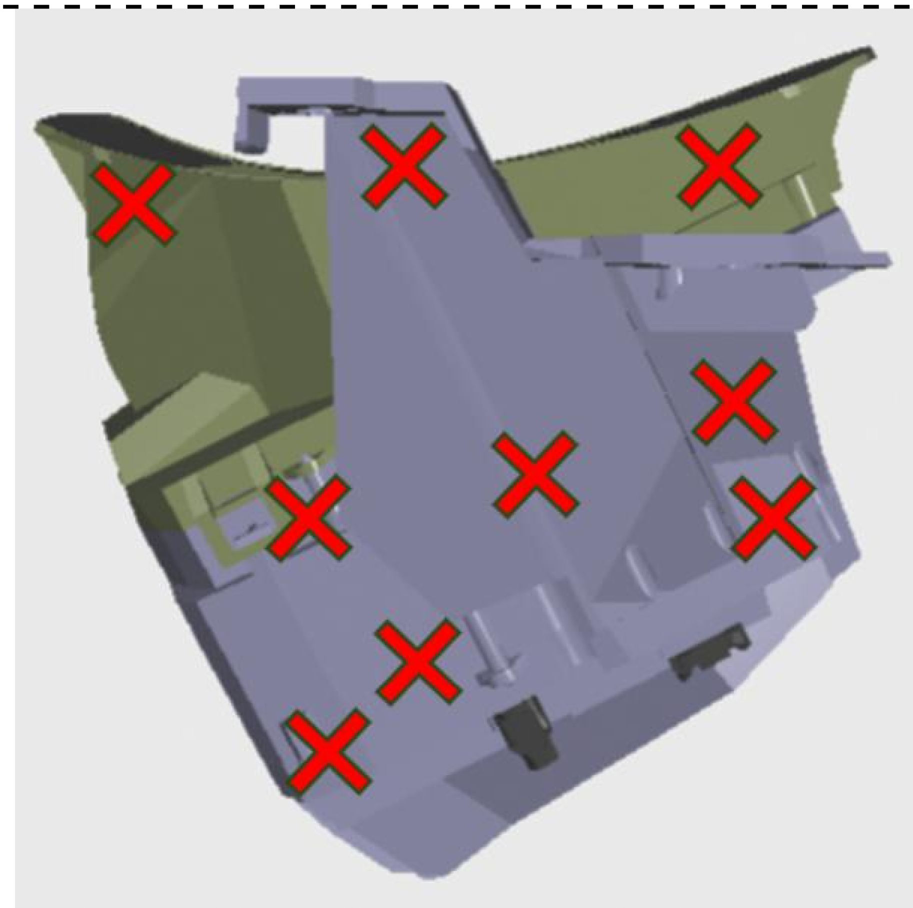
Bottom side



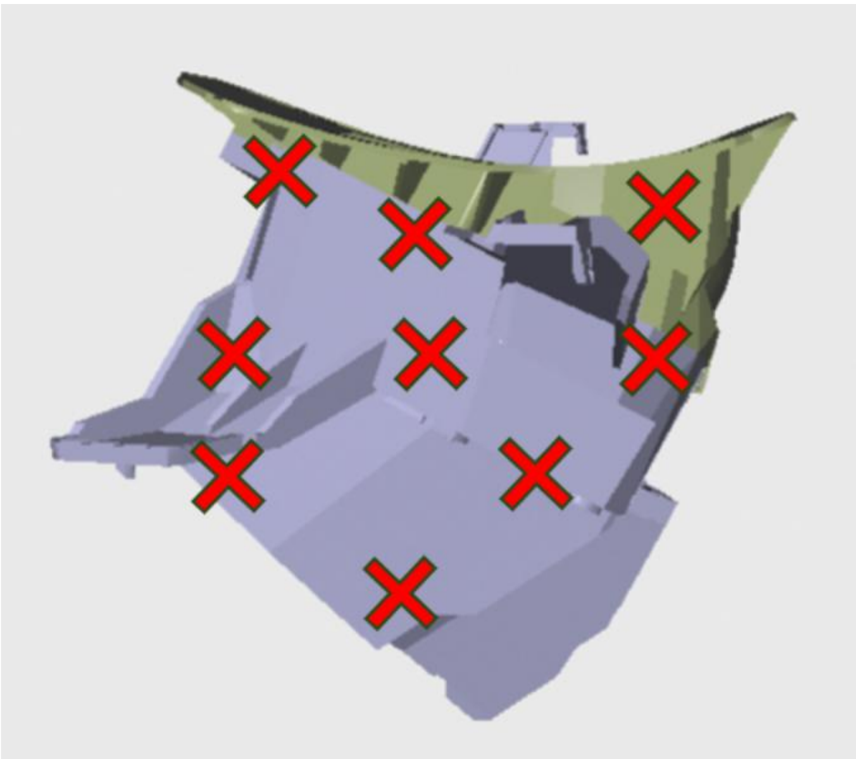
Top Side


Front Side


Back Side



Left Side



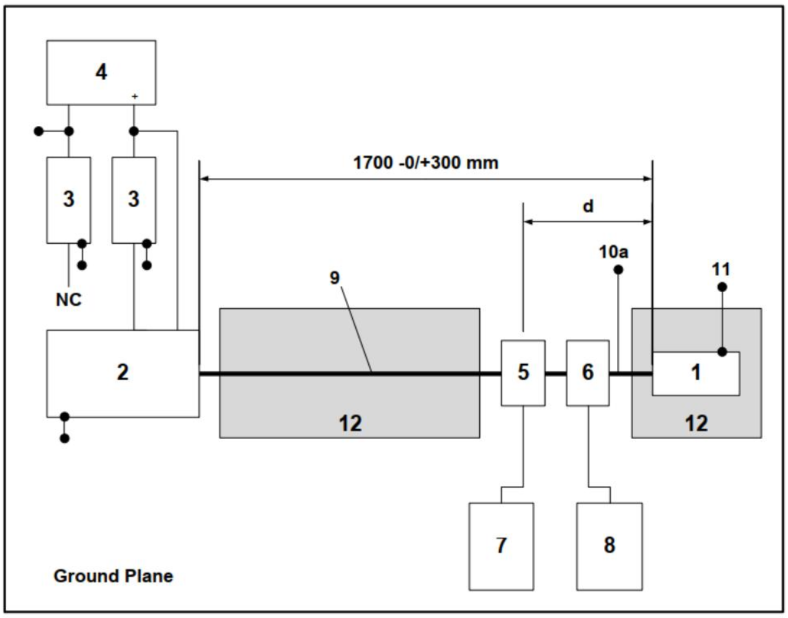
Right Side

Red points represent both air and contact discharges

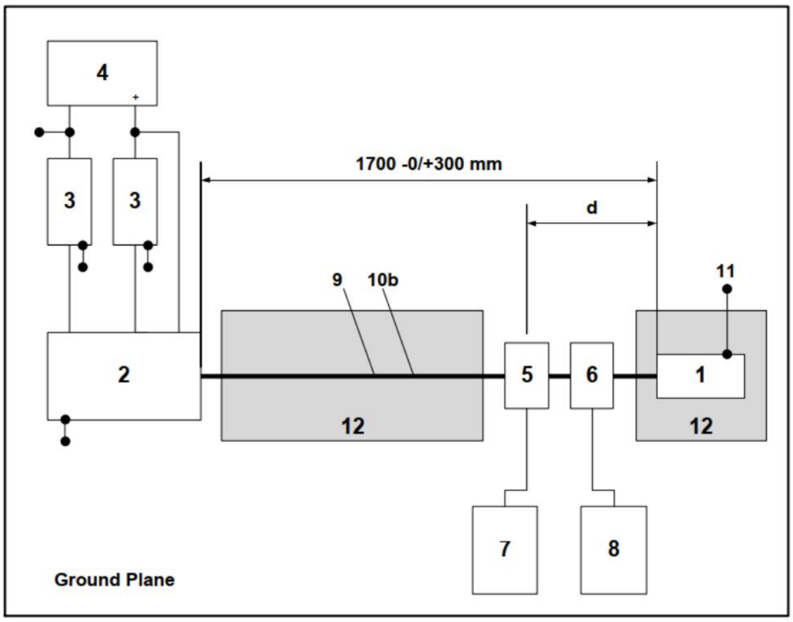
ESD test point

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 20.0 (powered) ISO 10605
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	See the ESD discharge points
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	/
DUT Monitoring Information	See the section 3.1

4.3 RF Immunity (RI 112)



DBCI



CBCI

Key:

1. DUT
2. Load Simulator
3. Artificial Network
4. Automotive Battery
5. Injection Probe
6. Monitor Probe (OPTIONAL)
7. RF Generation Equipment
8. Current Monitoring Equipment (OPTIONAL)
9. DUT Wire harness

10a. DUT Power Return removed from wire harness and connected directly to sheet metal. Wire length is 200 ±50 mm.

10b. DUT Power Return included in DUT wire harness

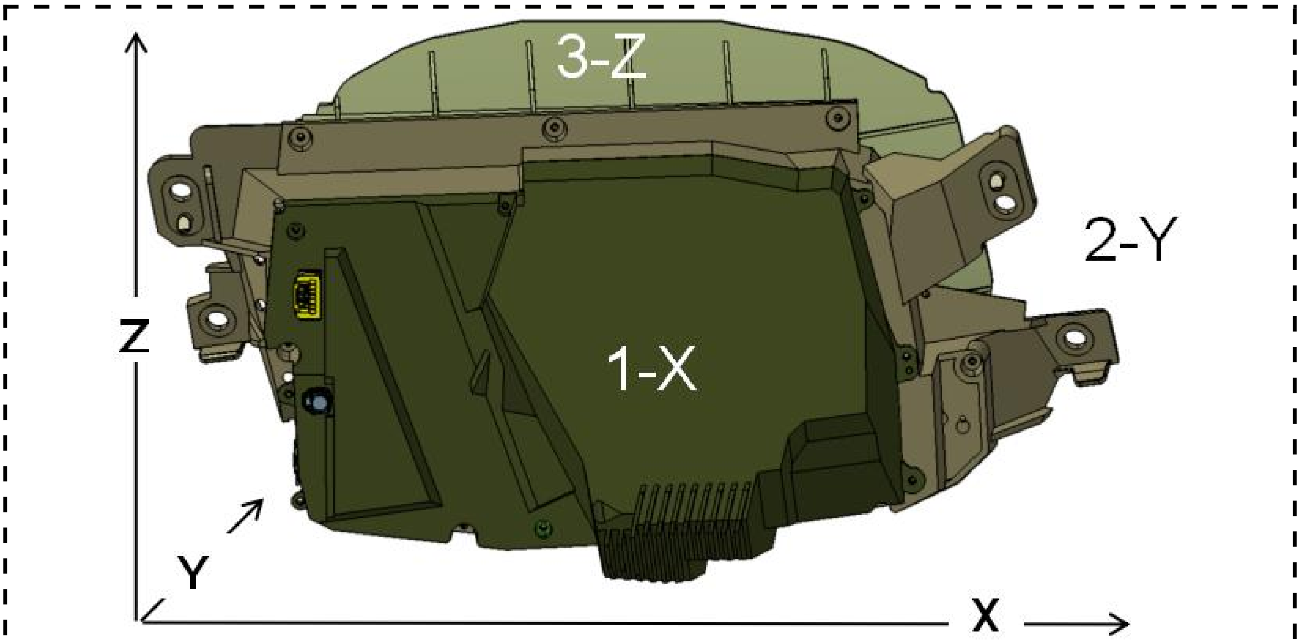
11. DUT Case Ground (where applicable)
12. Insulated Support ($\varepsilon_r \leq 1.4$)

d = Injection probe distance from DUT

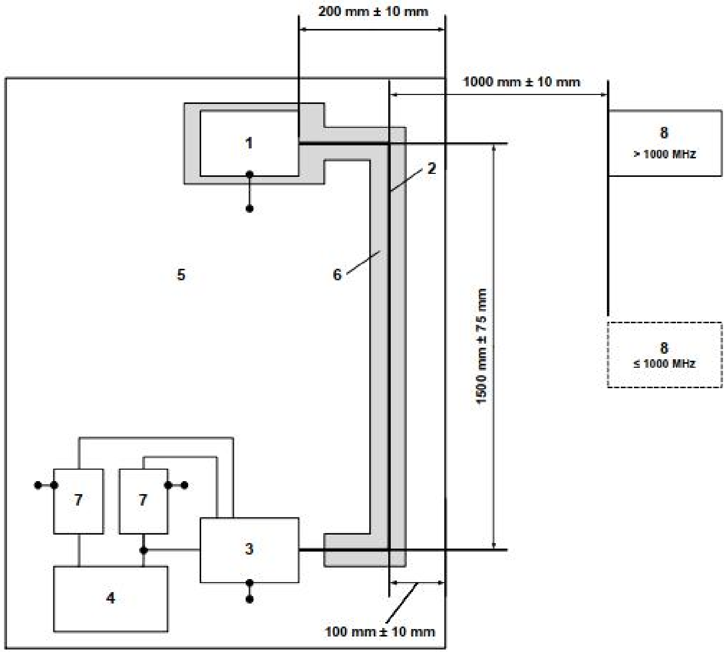
RI 112 Test Set-up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 10.3 ISO 11452-4
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	1. Input-Digital line only measure CBCI 2. Test each connector loom (harness) all wires from the connector included in the probe and then to test all connector harness (full bundle) together.
DUT Monitoring Information	See the section 3.1, during the test monitoring camera is aim at the TFT screen of the HUD to observe the image quality in the display and evaluate it in a visually subjective manner.

4.4 RF Immunity (RI 114)



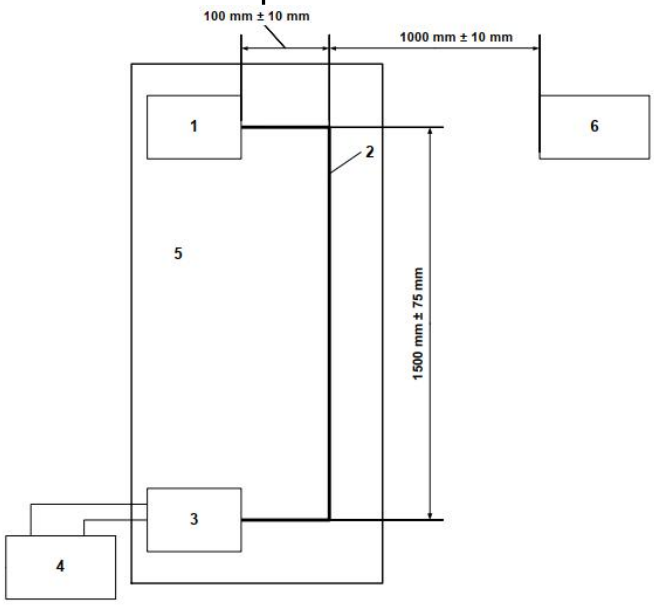
RI 114 Test surface



Key:

1. DUT
2. Wire harness
3. Load Simulator
4. Automotive Battery
5. Ground plane (bonded to shielded enclosure)
6. Insulated support ($\varepsilon_r \leq 1.4$)
7. Artificial Network
8. Transmitting Antenna

RI 114 ALSE Test Setup Bands 4 and 5



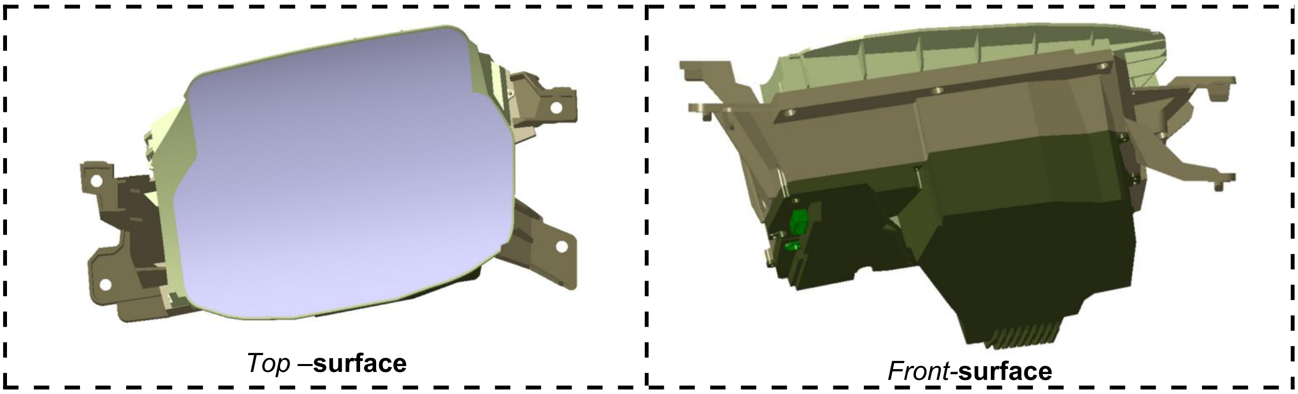
Key:

1. DUT
2. Wire harness
3. Load Simulator
4. Automotive Battery
5. Dielectric Support (($\varepsilon_r \leq 1.4$)
6. Transmitting Antenna

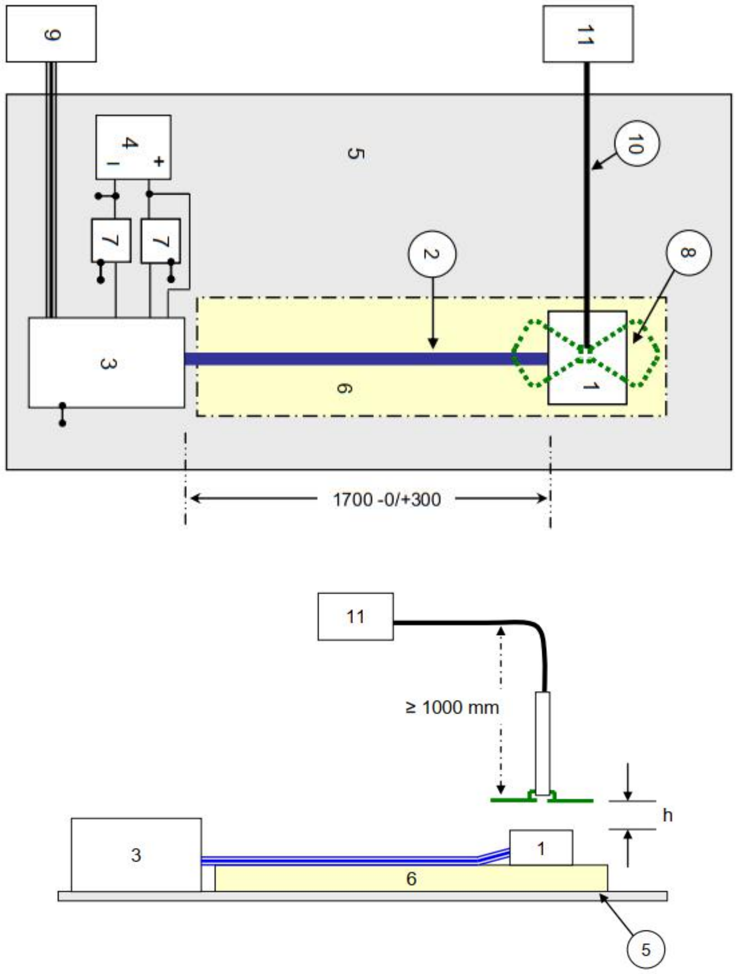
RI 114 ALSE Test Setup for Bands 6 and 7

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 10.4 ISO 11452-2
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm The RE 310 test harness shall be used to ensure consistency in the DUT and Load Simulator location
DUT Orientation	The connector is in the X direction
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	1. At ≥ 1000 MHz, the DUT shall be tested in a minimum of three (3) orthogonal orientations 2. Without 600V/m
DUT Monitoring Information	See the section 3.1

4.5 RF Immunity to Portable Transmitters (RI 115)



RI 115 Test Surface



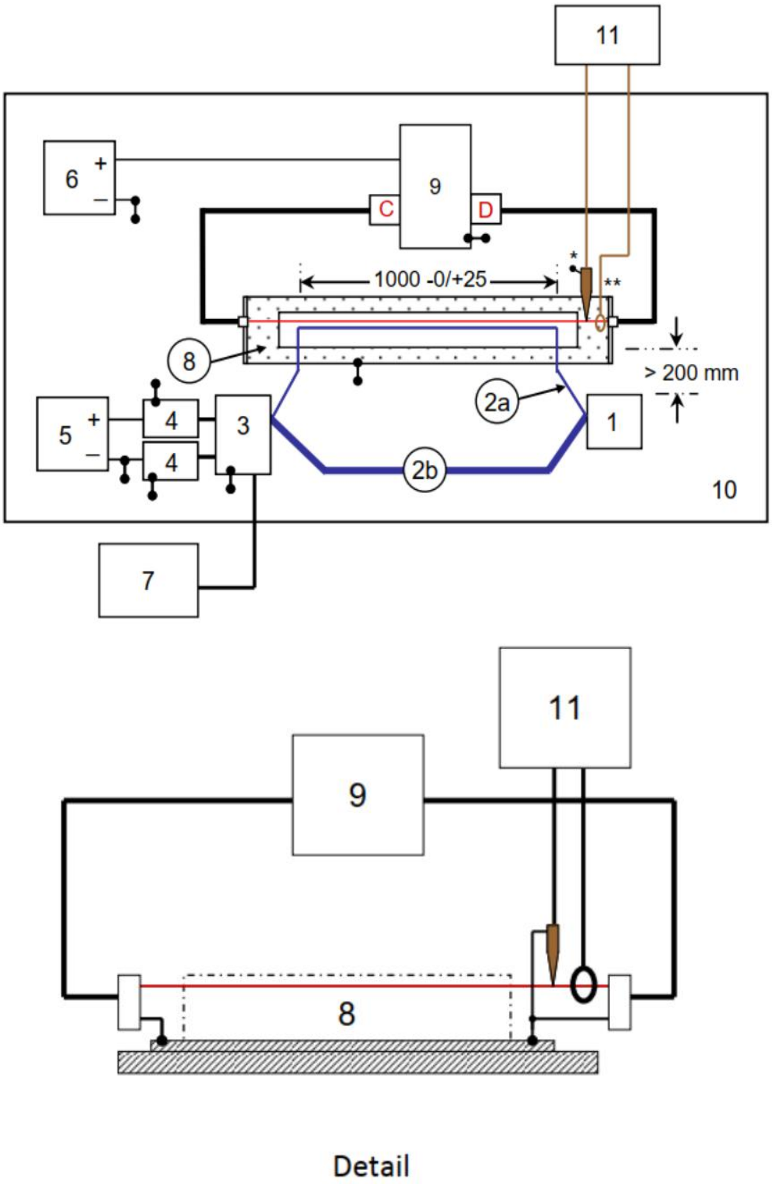
Key:

1. DUT
2. Test harness
3. Load Simulator
4. Automotive Battery
5. Ground Plane
6. Dielectric Support ($\varepsilon_r \leq 1.4$)
7. Artificial network
8. Test antenna (Schwarzbeck antenna SBA9113 with elements 420NJ)
9. Support Equipment
10. High quality double-shielded coaxial cable (cable can be no closer than 1000 mm to antenna elements. Place ferrite beads on cable (see Figure 10-6).
11. RF Generation Equipment (see Figure 10-6)

RI 115 Test Set-Up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 10.5 ISO 11452-9
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	See the figure RI 115 Test surface
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Antenna distance from DUT:50mm Antenna positioning steps:100mm
DUT Monitoring Information	See the section 3.1

4.6 Coupled Transient Noise (RI 130)



Key:

1	DUT
2a	DUT Circuit Wire to be tested
2b	DUT Wire Harness
3	Load Simulator
4	Artificial Network
5	Automotive Battery or linear power supply (Powers DUT and Load Simulator)
6	Automotive Battery (Powers Transient Test Generator)
7	DUT Monitor/Support Equipment
8	Coupling Test Fixture
9	Transient Generator (see Annex E for details). Generator connected to Coupling test fixture via coaxial cable. Case of generator connected to the ground plane.
10	Ground Plane

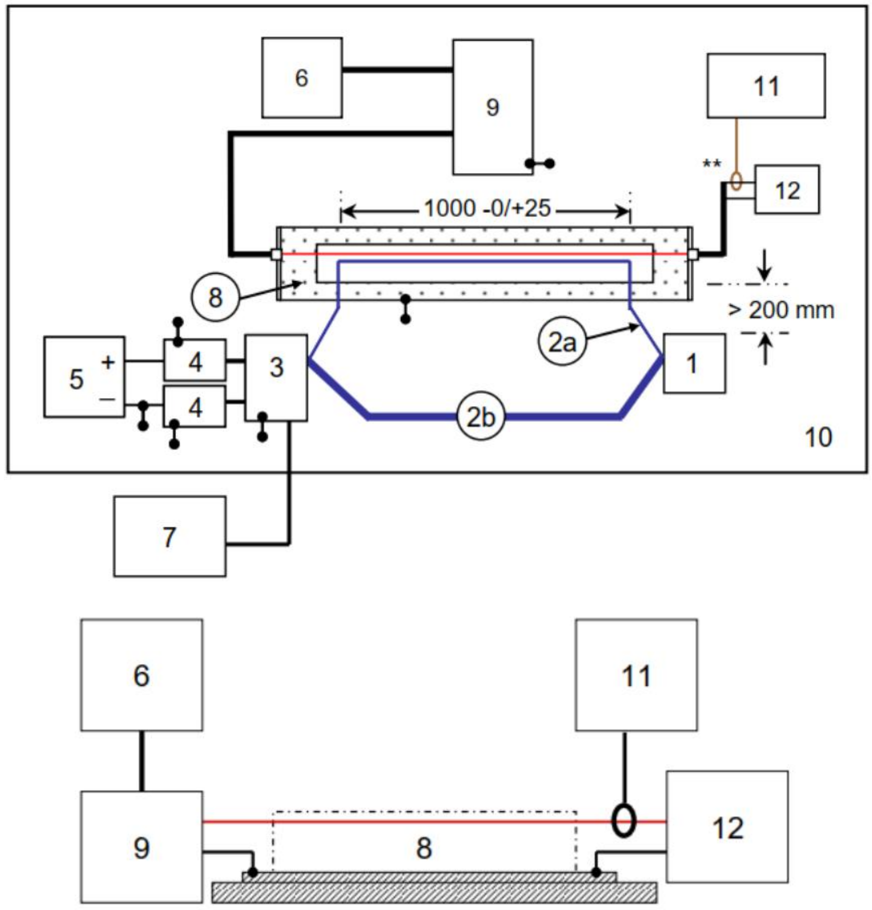
RI 130 Test Set-Up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 13
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	/
DUT Monitoring Information	See the section 3.1

4.7 Magnetic Field (RI 140)

N/A

4.8 Coupled Transient Noise (RI 150)



Detail

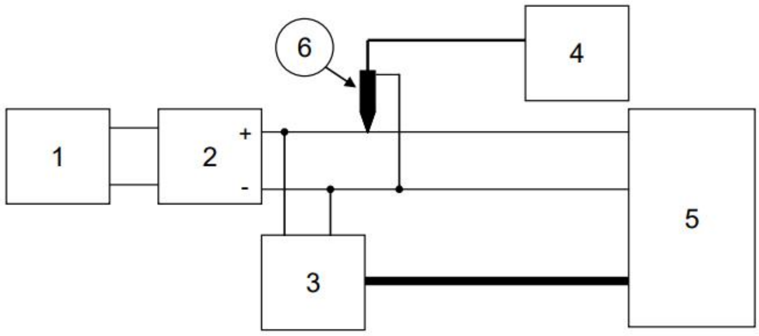
Key:

1	DUT
2a	DUT Circuit Wire to be tested
2b	DUT Wire Harness
3	Load Simulator
4	Artificial Network
5	Automotive Battery or linear power supply (Powers DUT and Load Simulator)
6	Signal Generator
7	DUT Monitor/Support Equipment
8	Coupling Test Fixture
9	Amplifier (1kHz – 100 kHz)
10	Ground Plane
11	Digital Oscilloscope (≥1 GS/sec, ≥ 8 Mega sample)
12	Amplifier Load Resistance (e.g. 4 ohms)

RI 150 Test Set-Up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 14
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	/
DUT Monitoring Information	See the section 3.1

4.9 Conducted Sine wave (CI 210)



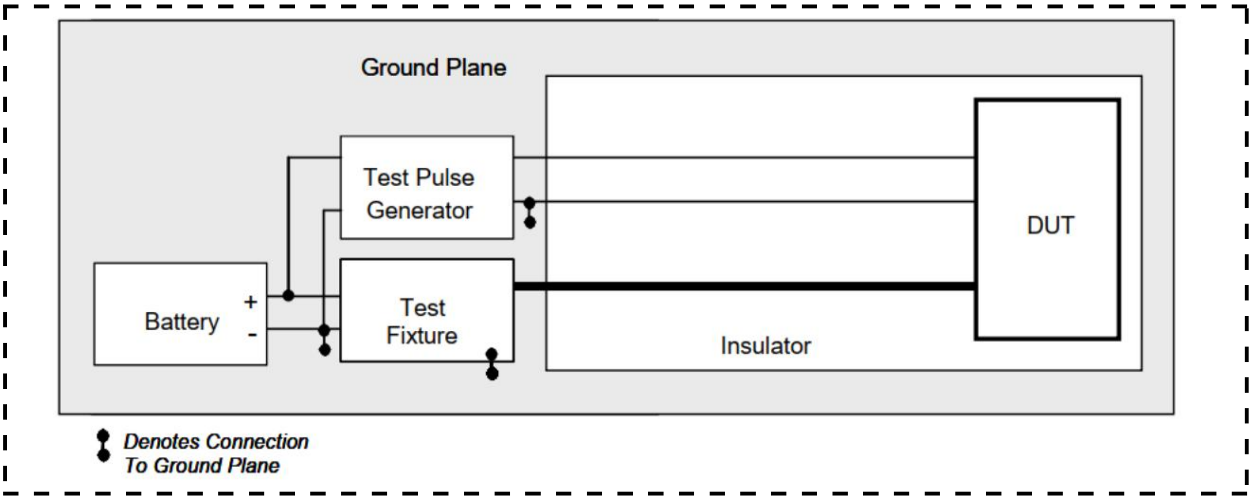
Key:

1. Signal Source
2. Modulated Power Supply (DC coupled Audio Amplifier with output impedance < 100 milliohms may be used)
3. Load Simulator
4. Oscilloscope
5. DUT
6. Passive High Impedance Probe (≥ 1Mohm, C< 10 pF)

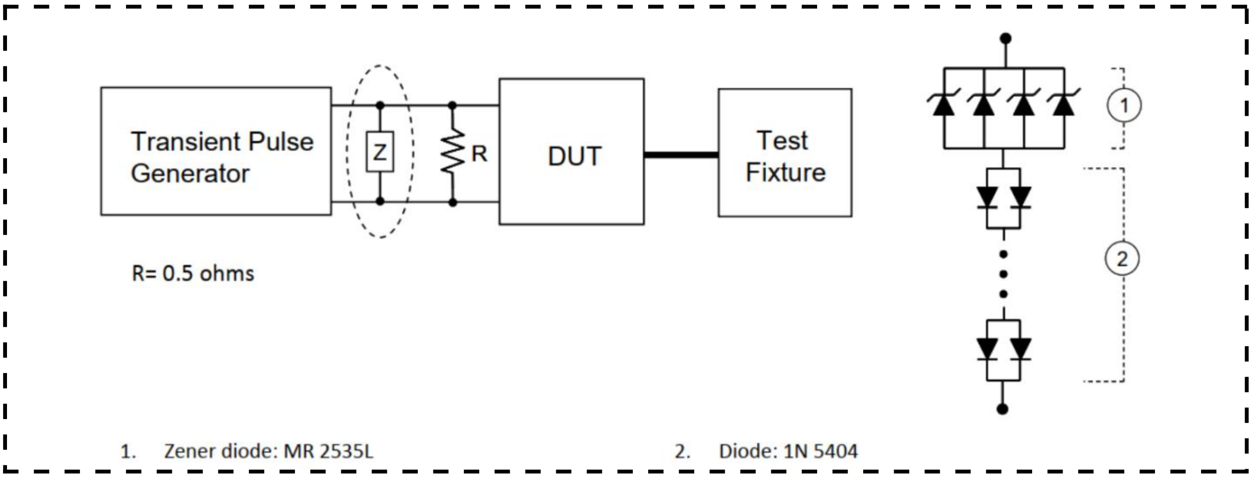
CI 210 Test Set-Up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 14.3
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	/
DUT Monitoring Information	See the section 3.1

4.10 Conducted Transients (CI 220)



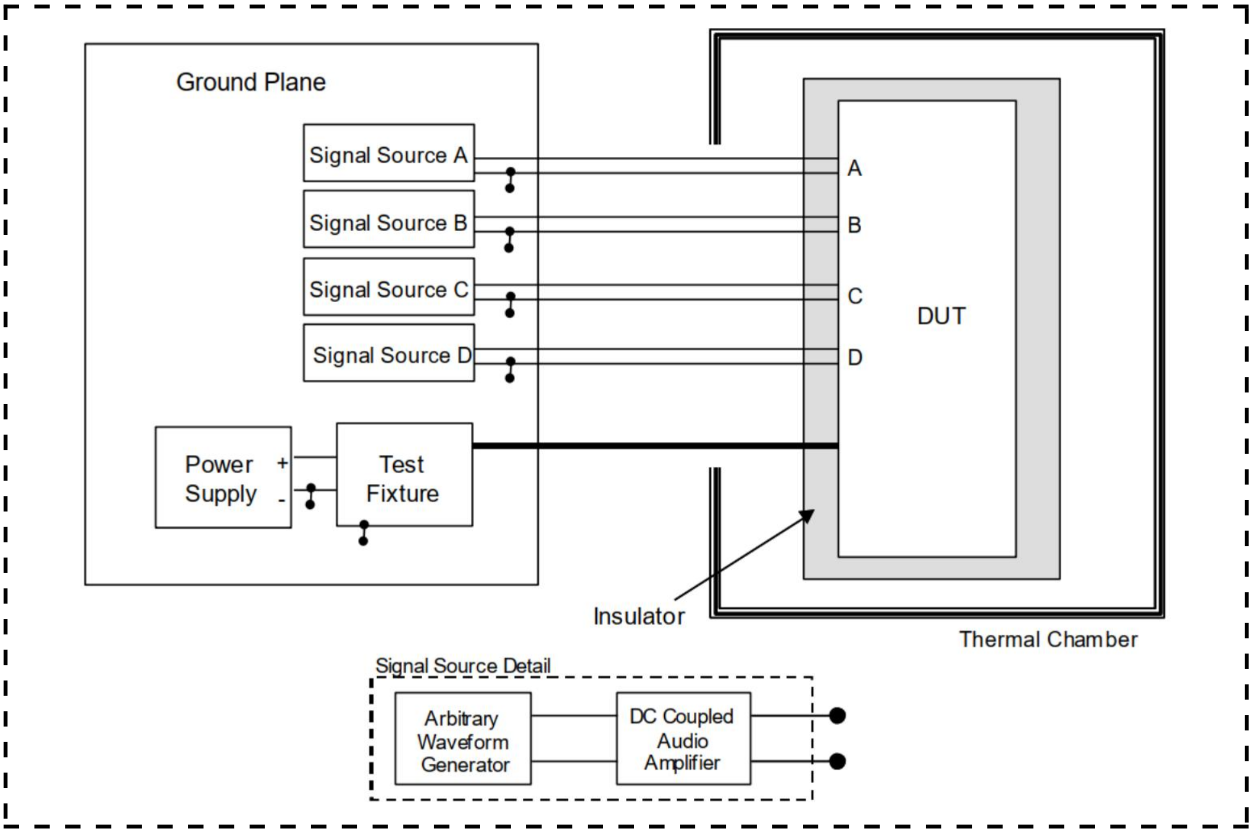
CI 220 Test Set-Up for Devices with a Single Power Supply Circuit



CI 220 Test setup for Application of Pulse G2

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 15
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	/
DUT Monitoring Information	See the section 3.1

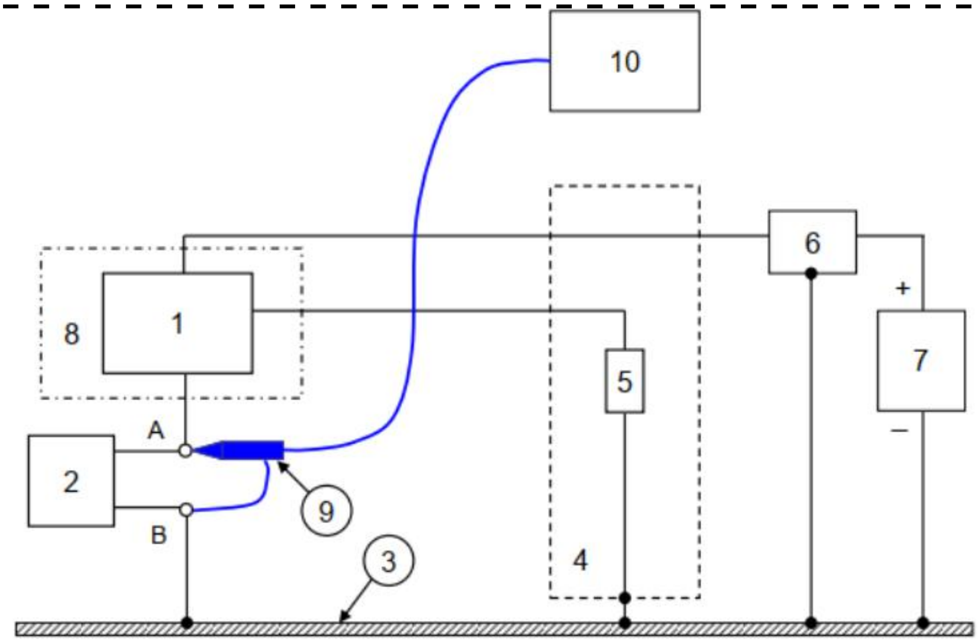
4.11 Power Cycling (CI 230)



CI 230 Power Cycling Test Setup

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 16
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Testing shall be performed at -40 +0 / - 5 ℃
DUT Monitoring Information	See the section 3.1

4.12 Ground Shift (CI 250)



a)

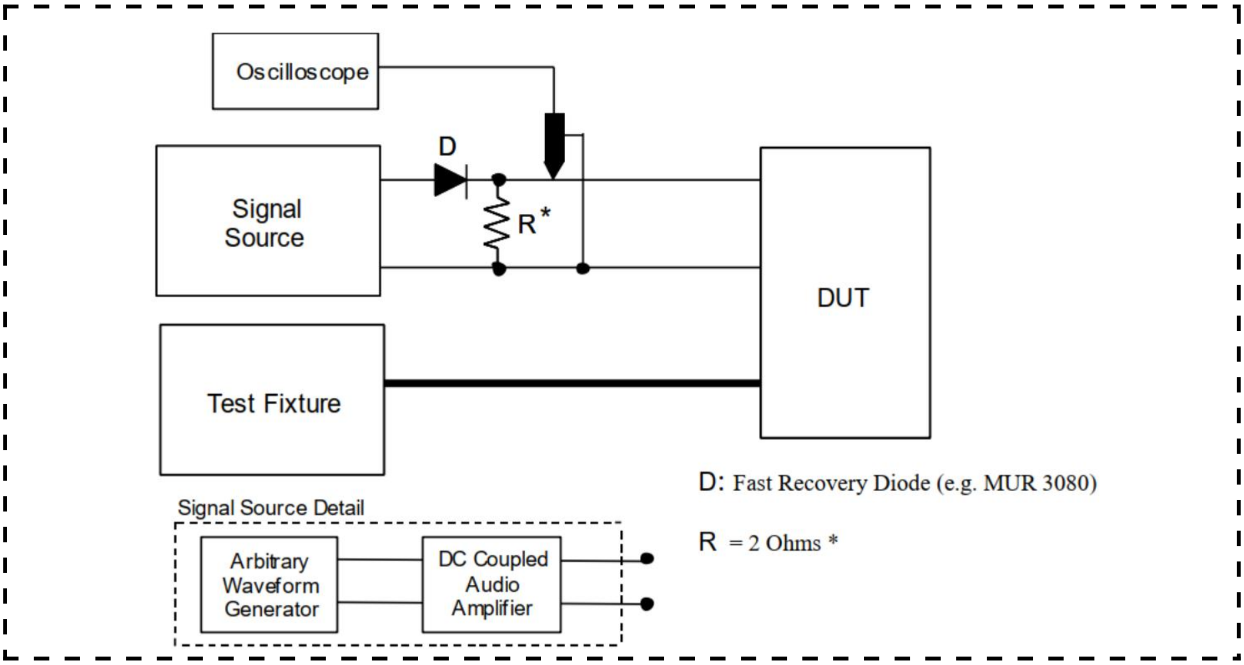
Key:

1. DUT
2. Signal Source (see Figure 17-5)
3. Ground Plane
4. Load Simulator
5. DUT External Load
6. Artificial Network
7. Power Supply (13.5 VDC)
8. Insulating Support
9. 10X high impedance probe (1M ohm, C< 10 pF)
10. Digital Oscilloscope (> 100 MS/s, > 6MB memory depth)

CI 250 Test Set-Up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 17
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	/
DUT Monitoring Information	See the section 3.1

4.13 Power Dropout (CI 265)



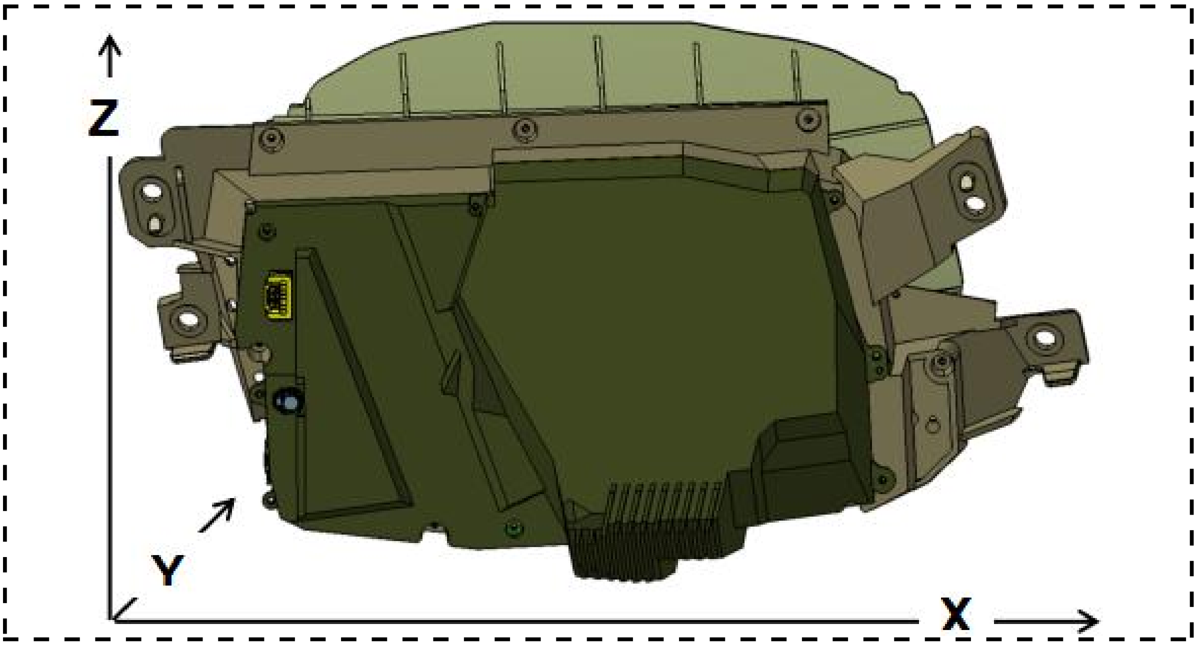
CI 265 Test Set-Up Detail for Waveform D

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 18
Deviations from Specified Test Method	
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Waveform A, B and C: The test harness connecting the DUT to the Test Fixture and transient pulse generator shall be ≤ 2000 mm in length
DUT Monitoring Information	See the section 3.1

4.14 DC Stress (CI 270)

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 19
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Test Set-Up refer to ‘unshielded room’ set-up diagrams in section 4.0
DUT Monitoring Information	See the section 3.1

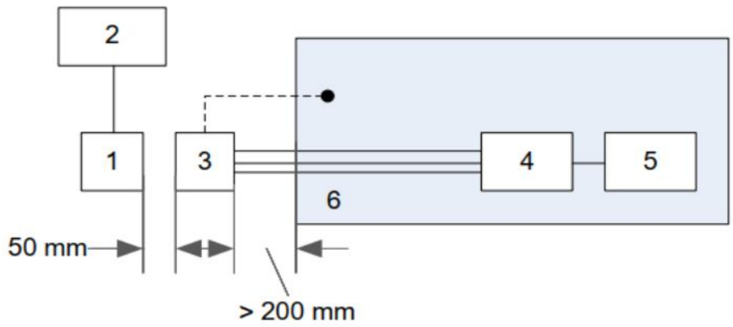
4.15 Radiated Emissions (RE 310)



RE 310 Test surface

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 7.0 CISPR 25
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	DUT connectors are facing the antenna
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Measurement receivers using Fast Fourier Transform (FFT) techniques may be used, however its use will be clearly stated in the EMC test report. For FFT measurements, a minimum of 10 continuous maximum hold scans shall be performed. If changes are observed between the scans, further tests shall be performed in groups of 5 scans until no perceptible changes in measured levels across the frequency range are observed. < 30 MHz: 1 m vertical monopole(Rod Antenna) 30~300 MHz: a biconical antenna; 200~1 000 MHz: a log-periodic antenna; 1 000~ 5 925 MHz: a horn or log periodic antenna From 150 kHz to 30 MHz measurements shall be performed in vertical polarisation only. From 30 MHz to 5 925 MHz measurements shall be performed in vertical and horizontal polarisations.
DUT Monitoring Information	See the section 3.1

4.16 Radiated Emissions (RE 320)



Configuration for Testing DUT only

Key:

1. Loop antenna
2. Measurement Receiver
3. DUT
4. Load Simulator
5. Artificial Network
6. Ground Plane

RE 320 Test Set-up

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 11 Defence Standard 59-411 Part 3 Issue 1 Amendment 1 DRE02.B
Deviations from Specified Test Method	/
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	Top--plan
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Measurement receivers using Fast Fourier Transform (FFT) techniques may be used, however its use will be clearly stated in the EMC test report. For FFT measurements, a minimum of 10 continuous maximum hold scans shall be performed. If changes are observed between the scans, further tests shall be performed in groups of 5 scans until no perceptible changes in measured levels across the frequency range are observed.
DUT Monitoring Information	See the section 3.1

4.17 Conducted Transient (CE 410)

Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 9.0 ISO 7637-2　4.3
Deviations from Specified Test Method	/
Harness Configuration	Power line and return line:200mm±50mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	DUT states:1. OFF to ON, 2. ON to OFF Test Set-Up refer to ISO 7637-2
DUT Monitoring Information	See the section 3.1

4.18 Conducted Emissions (CE 420)

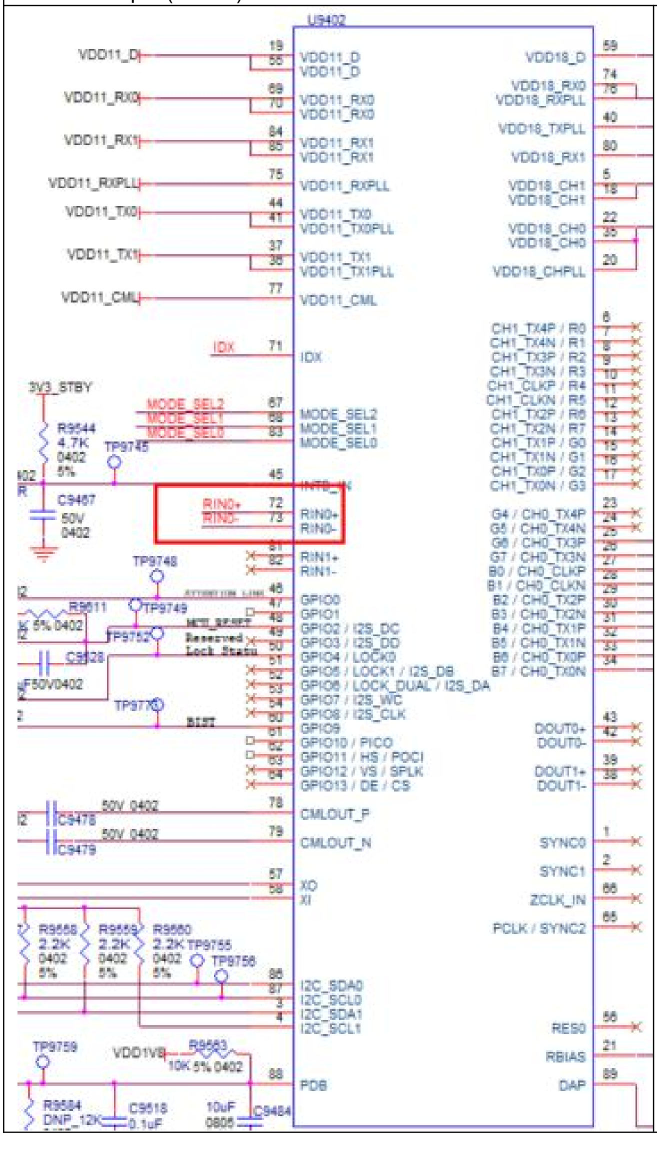
Test Details	Comments
Specified Test Method	Standard JLR-EMC-CS v1.0, Section 10.4
Deviations from Specified Test Method	Without CP-03
Harness Configuration	Signal lines have shielded twisted-pair All harness lengths：1700+300mm/-0mm
DUT Orientation	/
DUT Grounding (case or harness)	Case Ground the case in accordance with f in section 1.4.
Additional Test Specific Information	Test Set-Up refer to IEC CISPR 25
DUT Monitoring Information	See the section 3.1

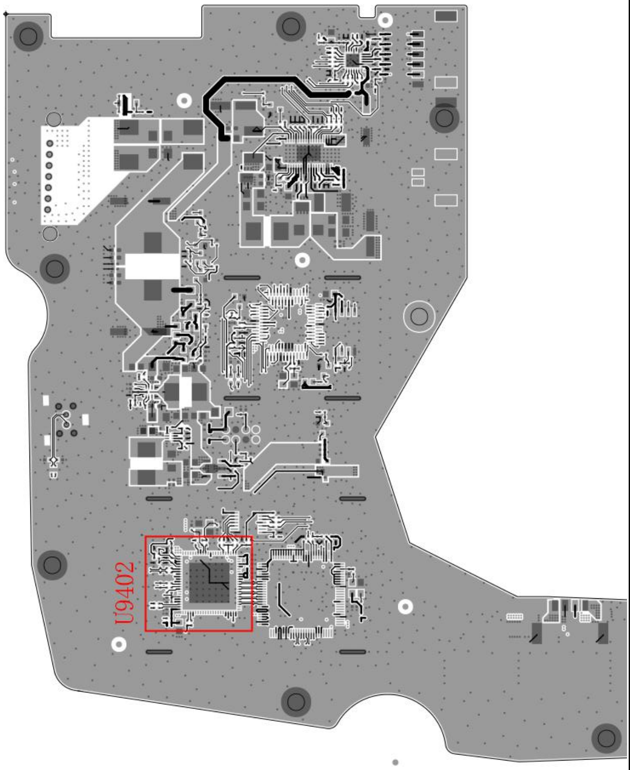
Appendix A--- Teardown Analysis

1. X-Ray

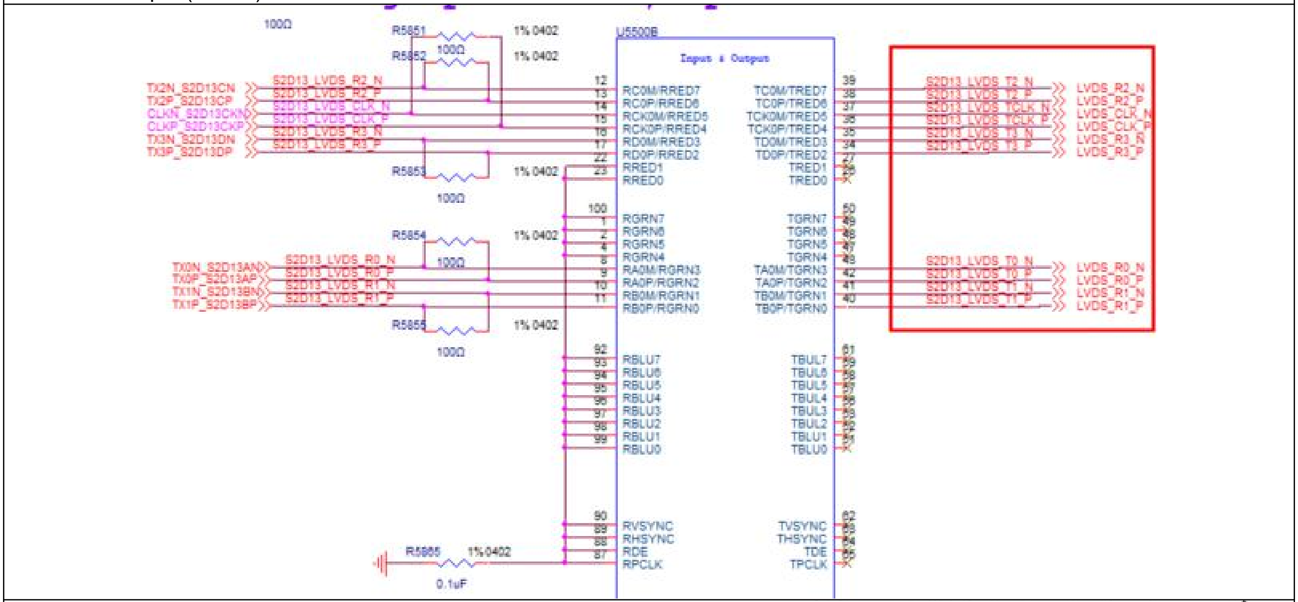
Test object	Inspection method	Test Site	Test Requirements	Remark
RIN0± input (U9402)	Measure the position corresponding to the input/ output of the critical component using X-Ray	COT	Solder joints: no virtual welding or short circuit. Connection line: no disconnection, short circuit and other abnormalities.	X-ray diagram attached to the teardown verification report
LVDS Output (U5500)				
MCU (U9406)				

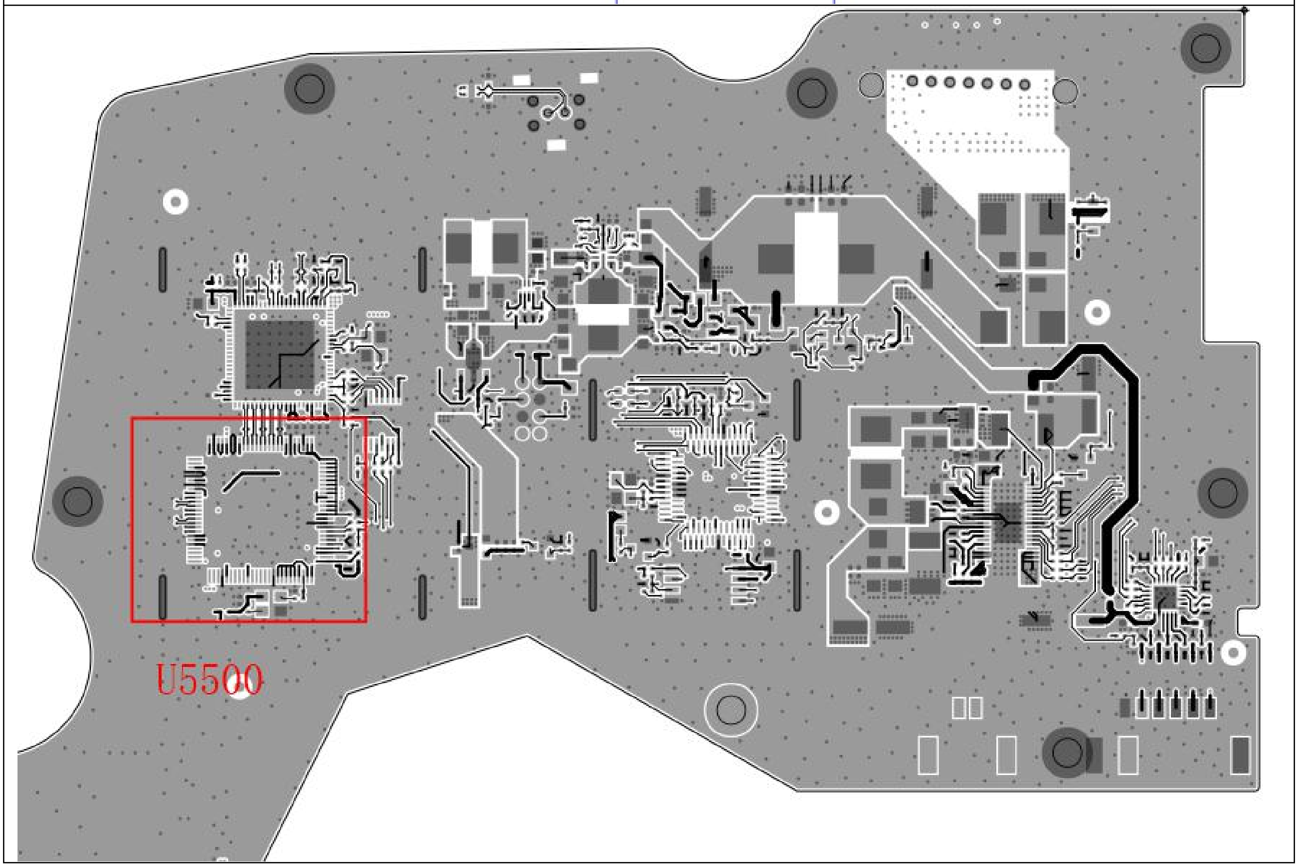
1.1 RIN0±input (U9402)：



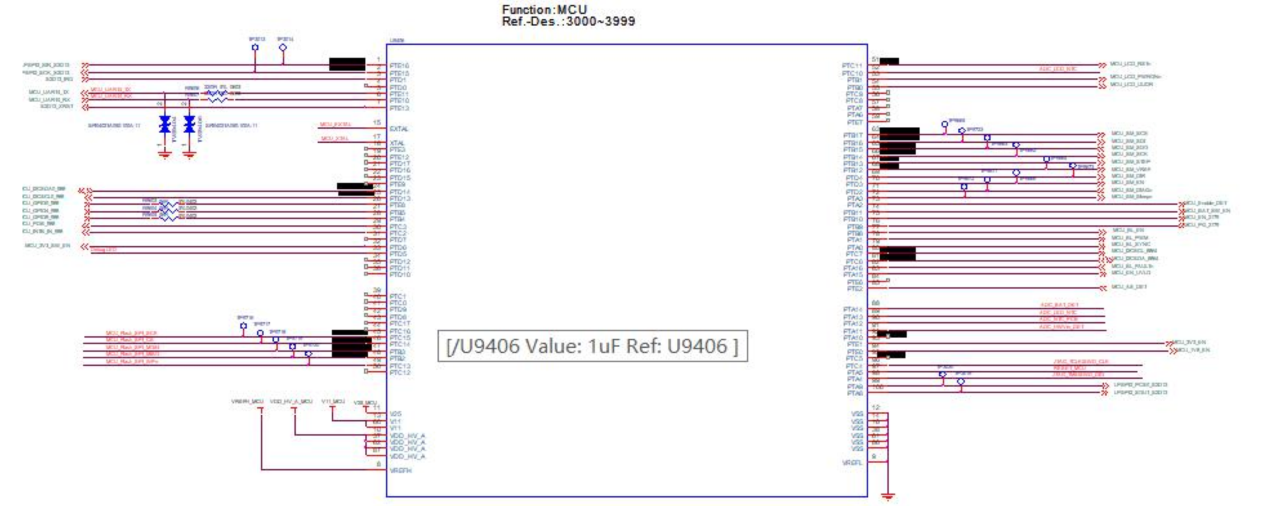


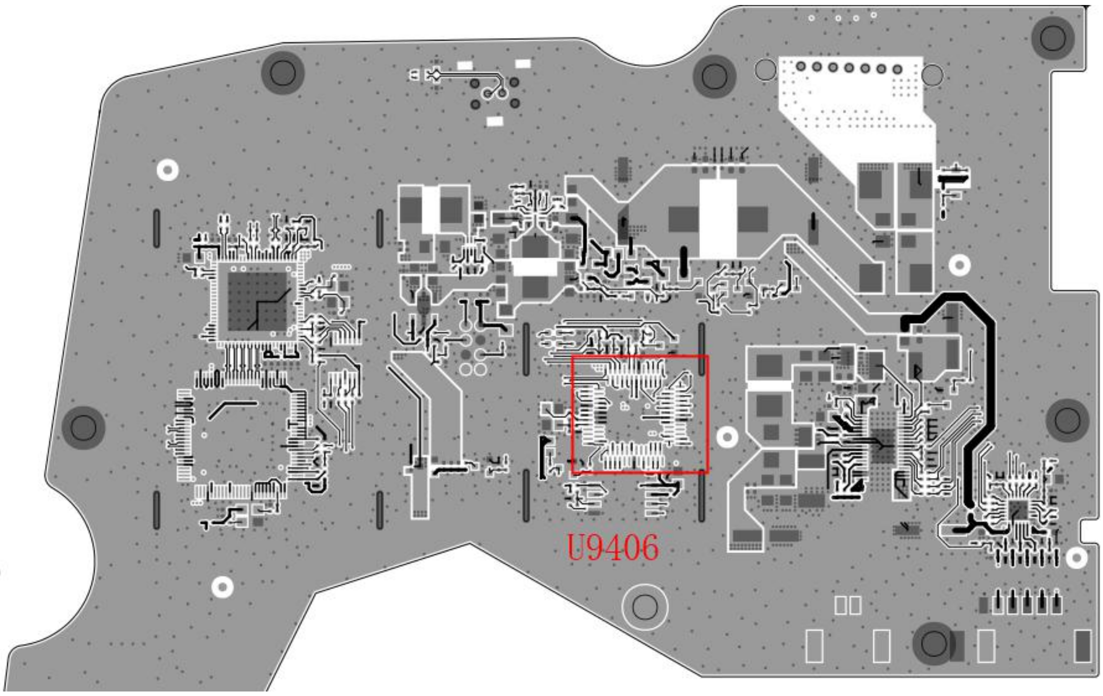
1.2 LVDS Output (U5500)：





1.3 MCU(U9406)

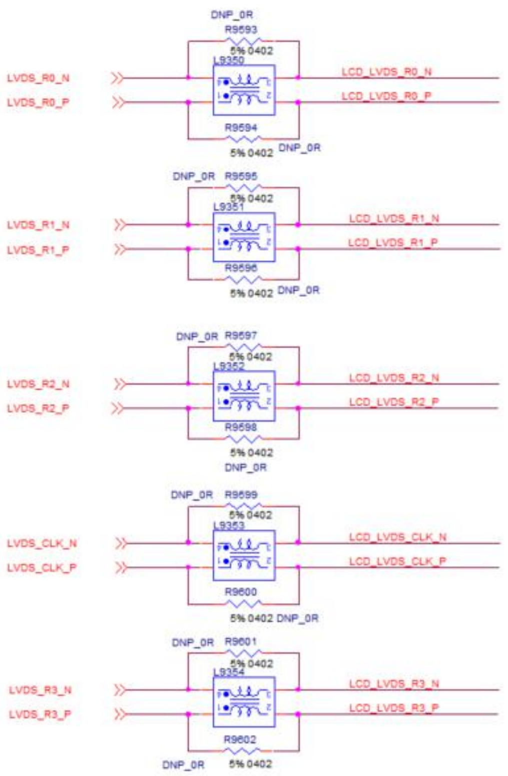

Function:MCU
Ref.-Des.:3000~3999
[/U9406 Value: 1uF Ref: U9406]

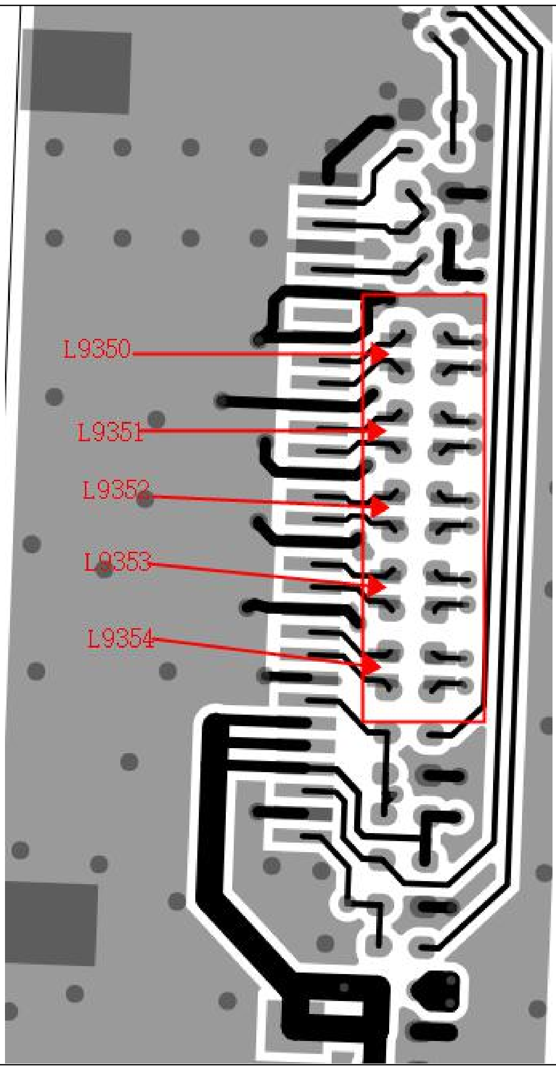

U9406

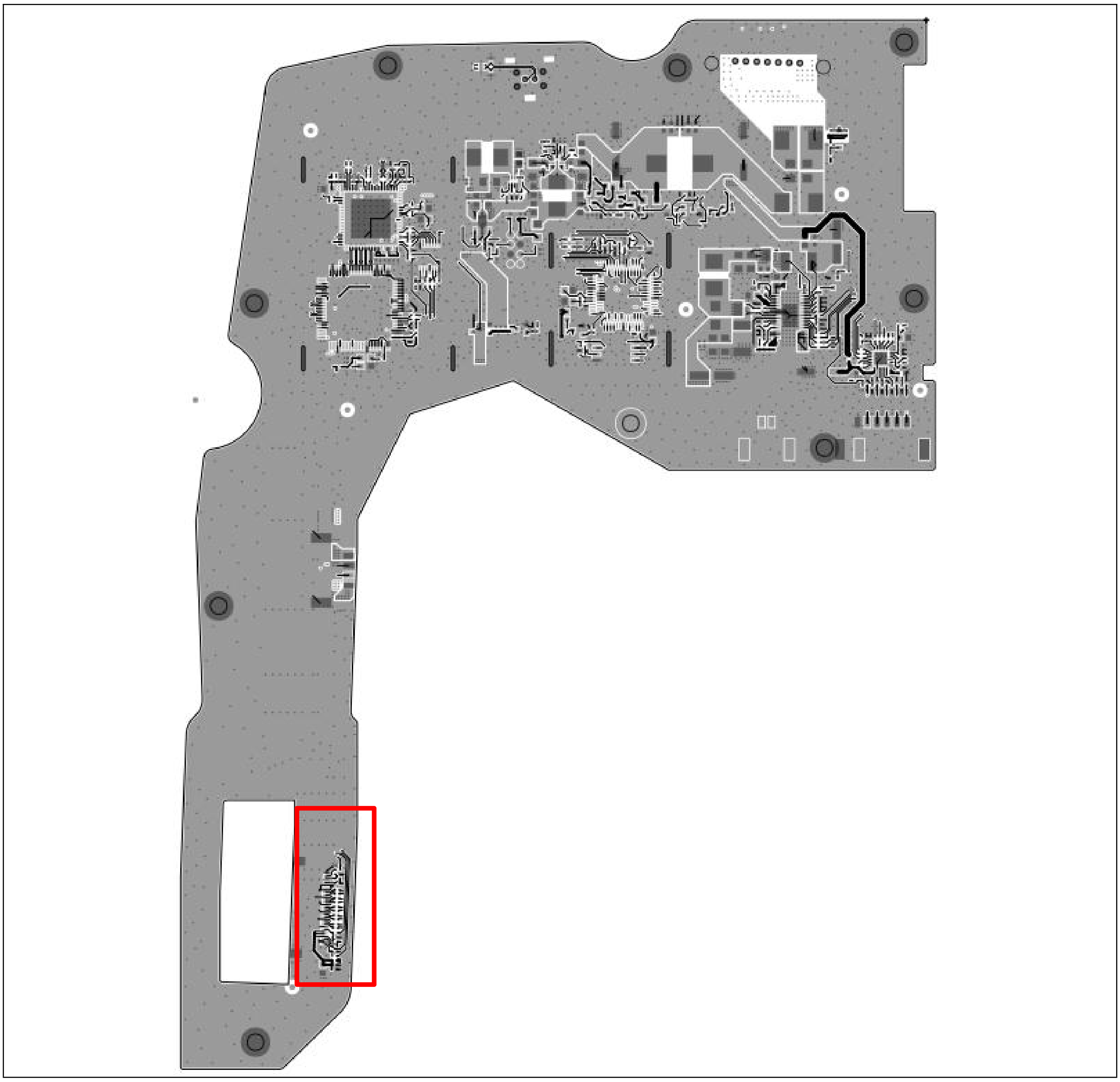
2. Measure

Test object	Inspection method	Test Site	Test Requirements	Remark
LVDS (L9350-L9354)	Single board measurement	COT	See section 2.1 below	Battery Voltage 13.5V
3.3V for MCU (C4012)			3.3V±0.15V	Battery Voltage 13.5V
1.15V for Deserializer (C9498)			1.15V±0.04V	Battery Voltage 13.5V
1.8V for Deserializer (C4201)			1.8V±0.11V	
Motor Driver (C8513)			13.5V±1.5V	Battery Voltage 13.5V
LED Driver (C9425)			±5%	Need to be tested at maximum brightness
TVS (TVS8001\ TVS8000\ TVS5250\ TVS5251)			See section 2.7below	

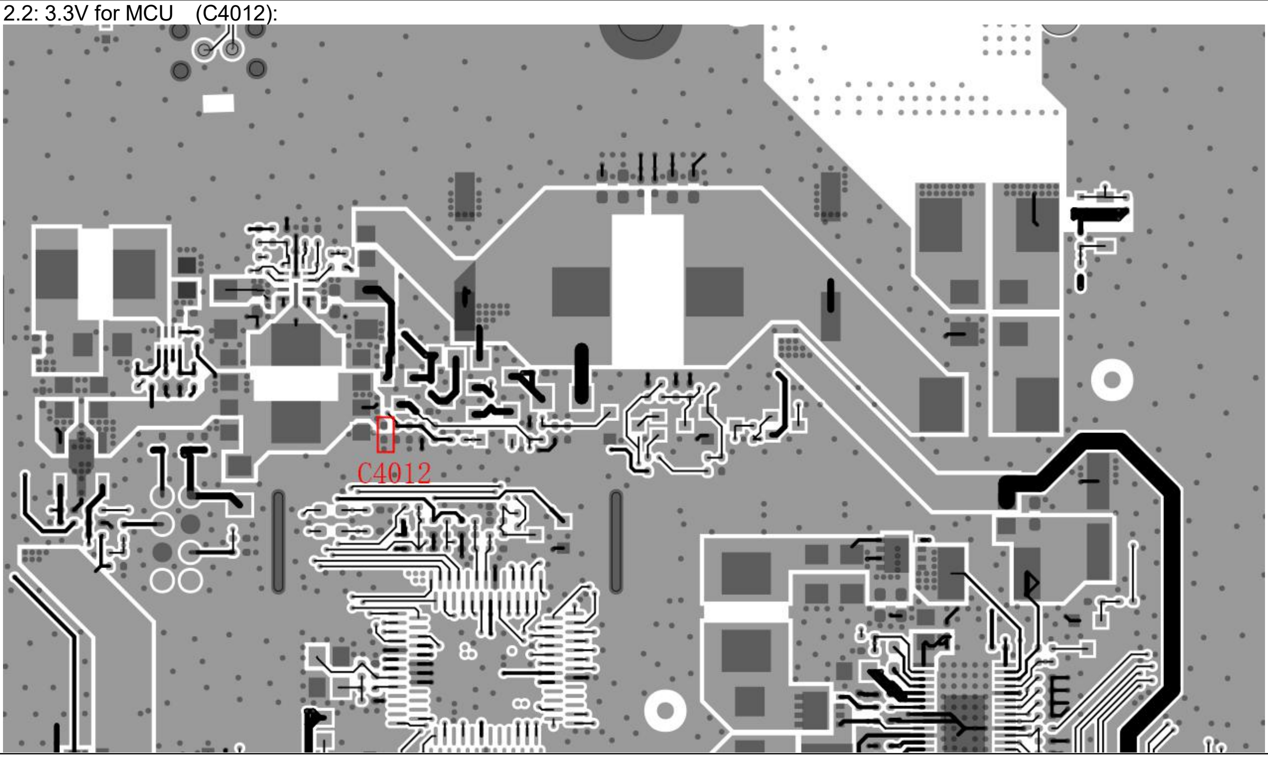
2.1 LVDS(L9350-L9354)：
LCD_LVDS_R0_N(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R0_P(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R1_N(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R1_P(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R2_N(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R2_P(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R3_N(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_R3_P(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_CLK_N(Nominal value 5.6MΩ,Tolerance±10%)
LCD_LVDS_CLK_P(Nominal value 5.6MΩ,Tolerance±10%)



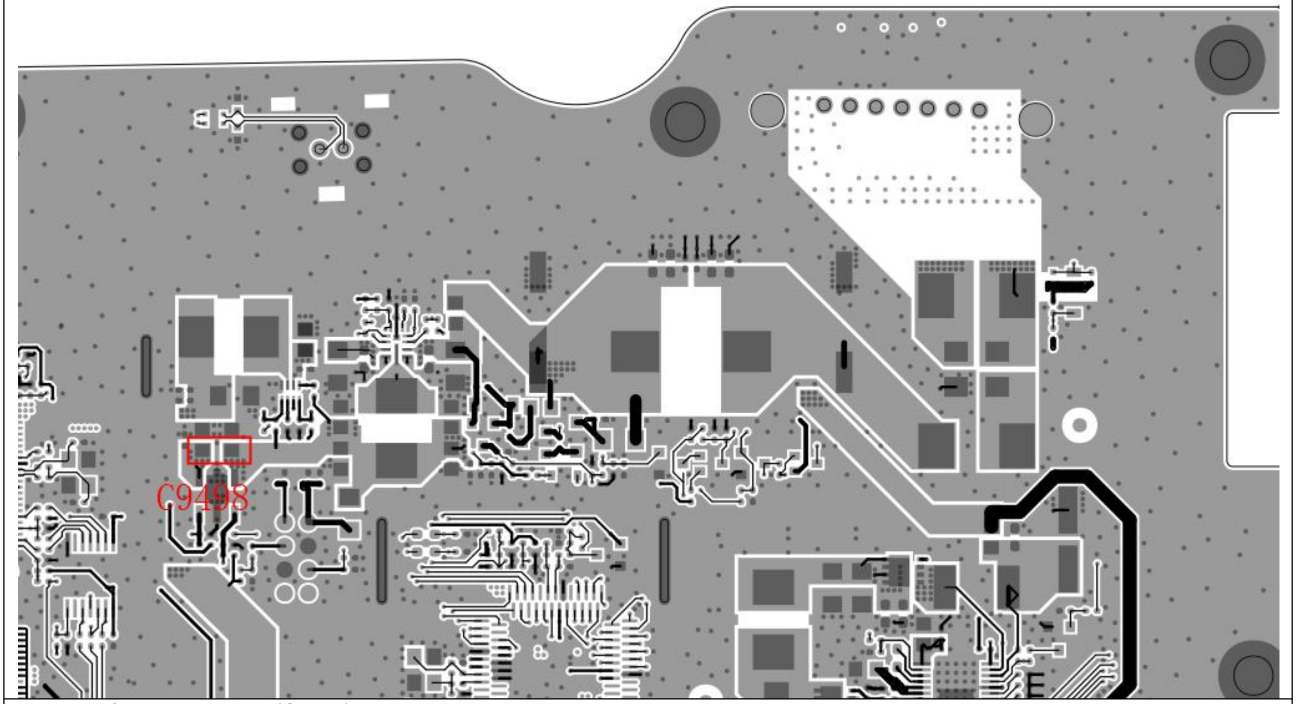




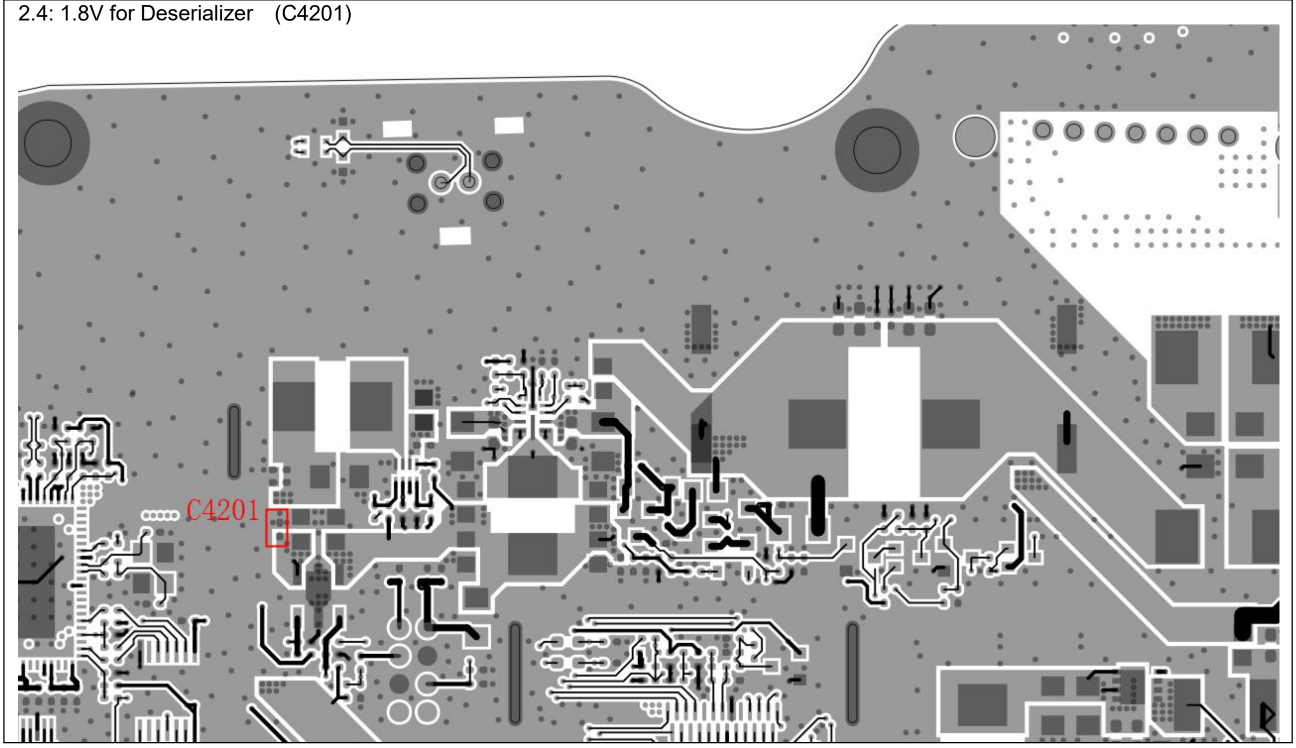
2.2: 3.3V for MCU (C4012):



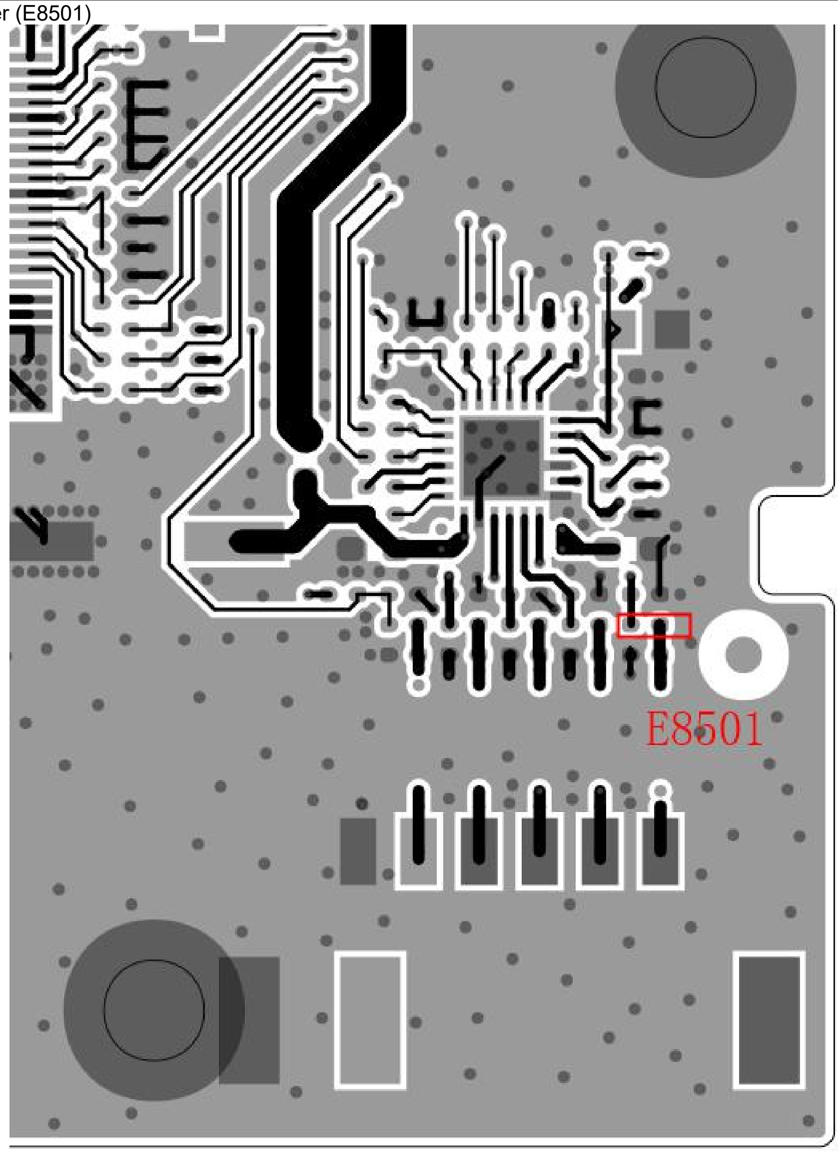
2.3: 1.15V for Deserializer (C9498):



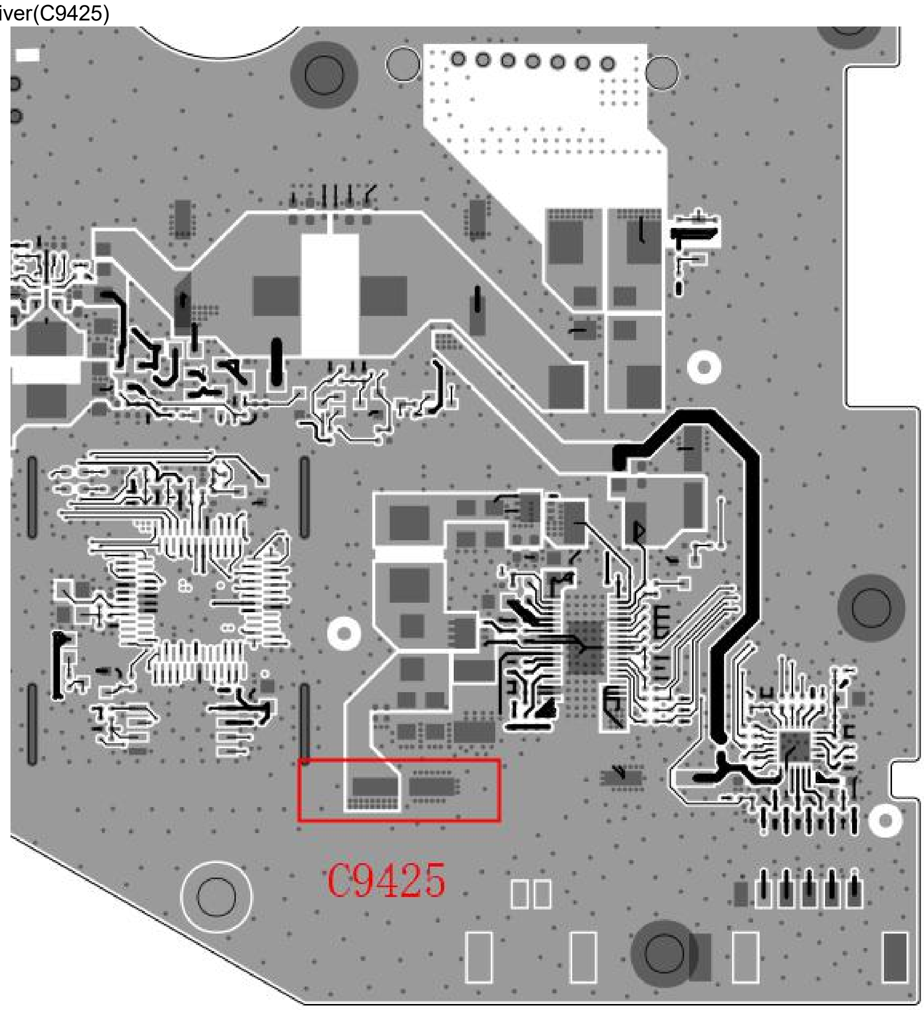
2.4: 1.8V for Deserializer (C4201)



2.5:Motor Driver (E8501)

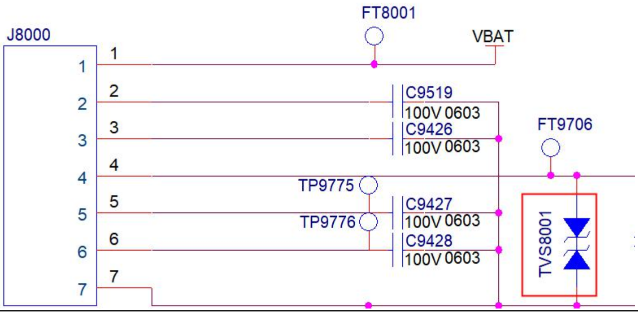


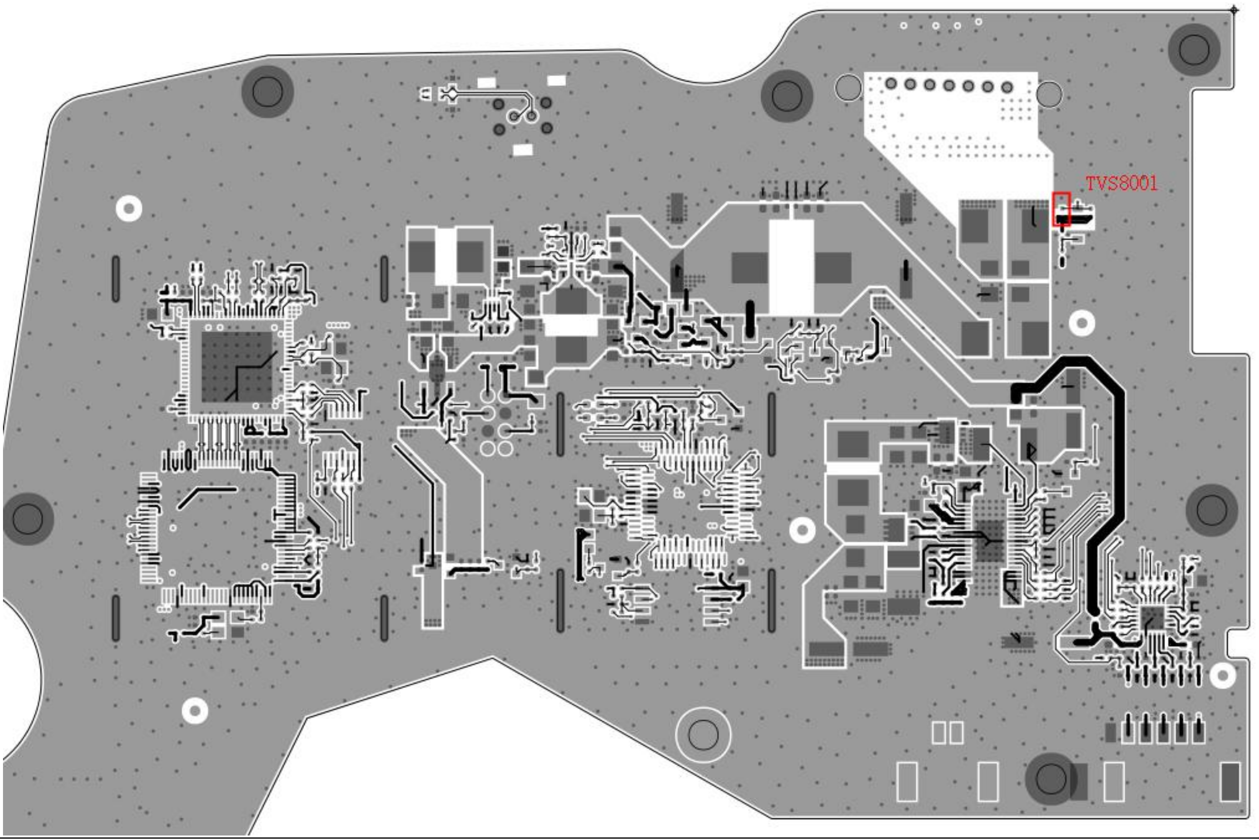
2.6：LED Driver(C9425)



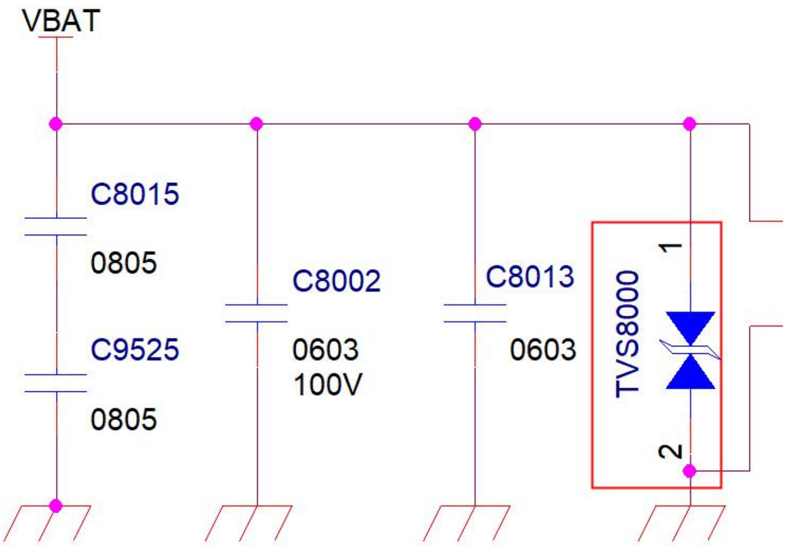
2.7 TVS(TVS8001\ TVS8000\ TVS5250\ TVS5251)
-TVS8001 for Enable(2.2nF, Tolerance ±25%)
-TVS8000 for VBAT(19MΩ, Tolerance ±10%)
-TVS5250 for RIN0+(100nF,Tolerance ±25%)
-TVS5251 for RIN0- (100nF,Tolerance ±25%)

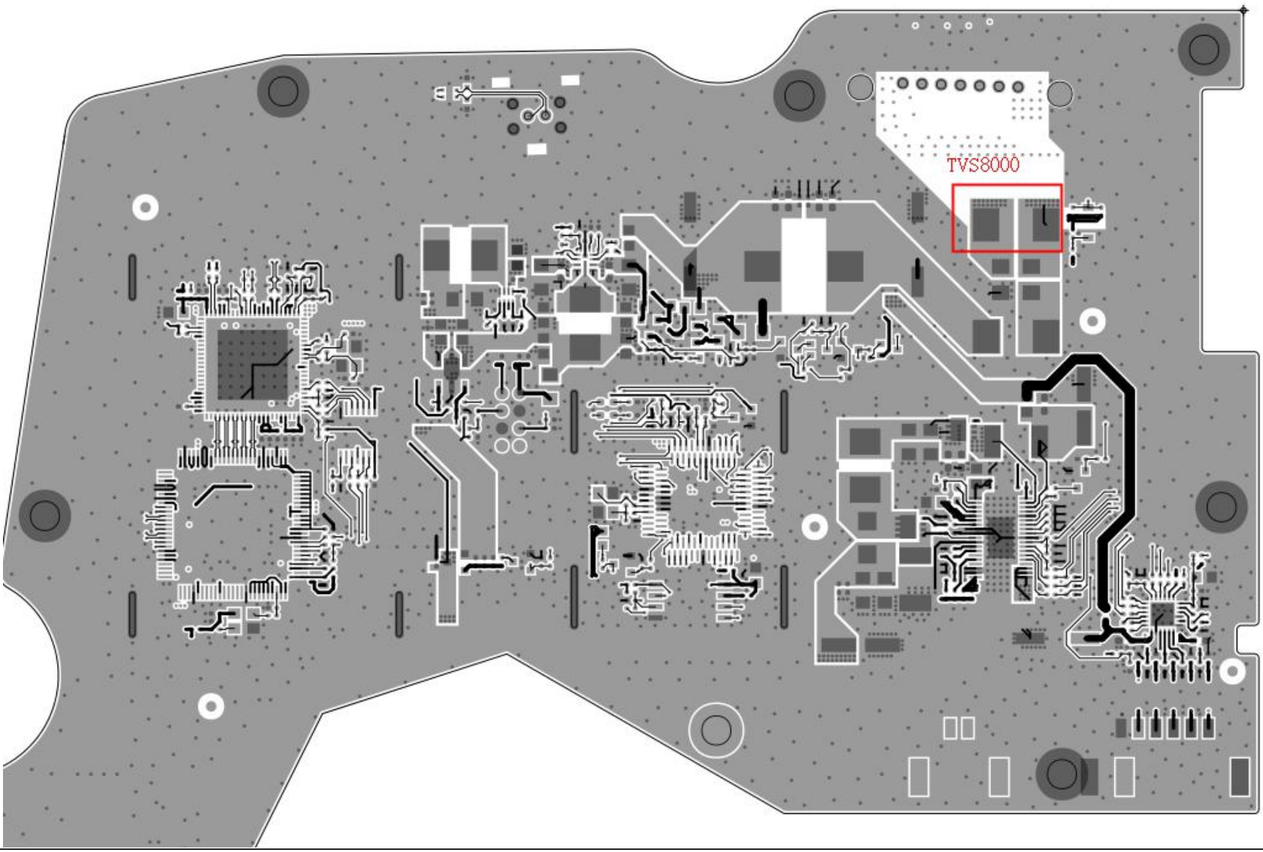
TVS8001 for Enable:



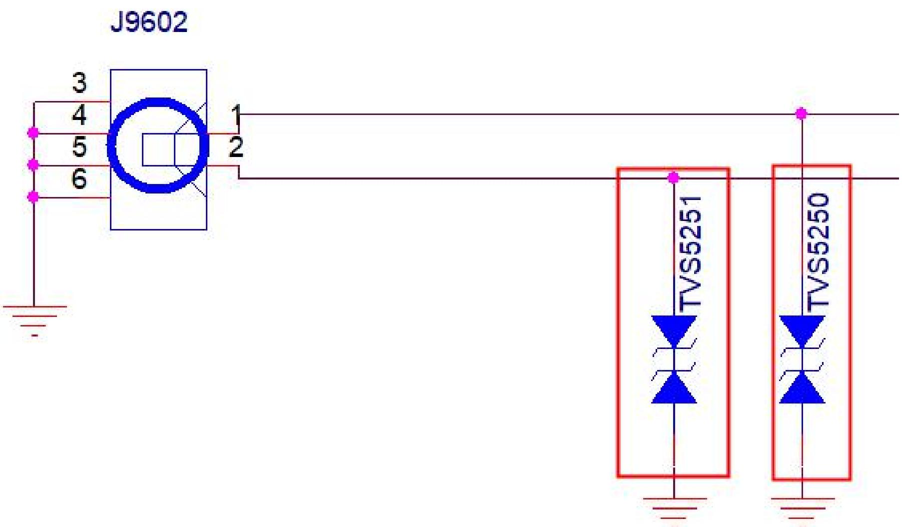


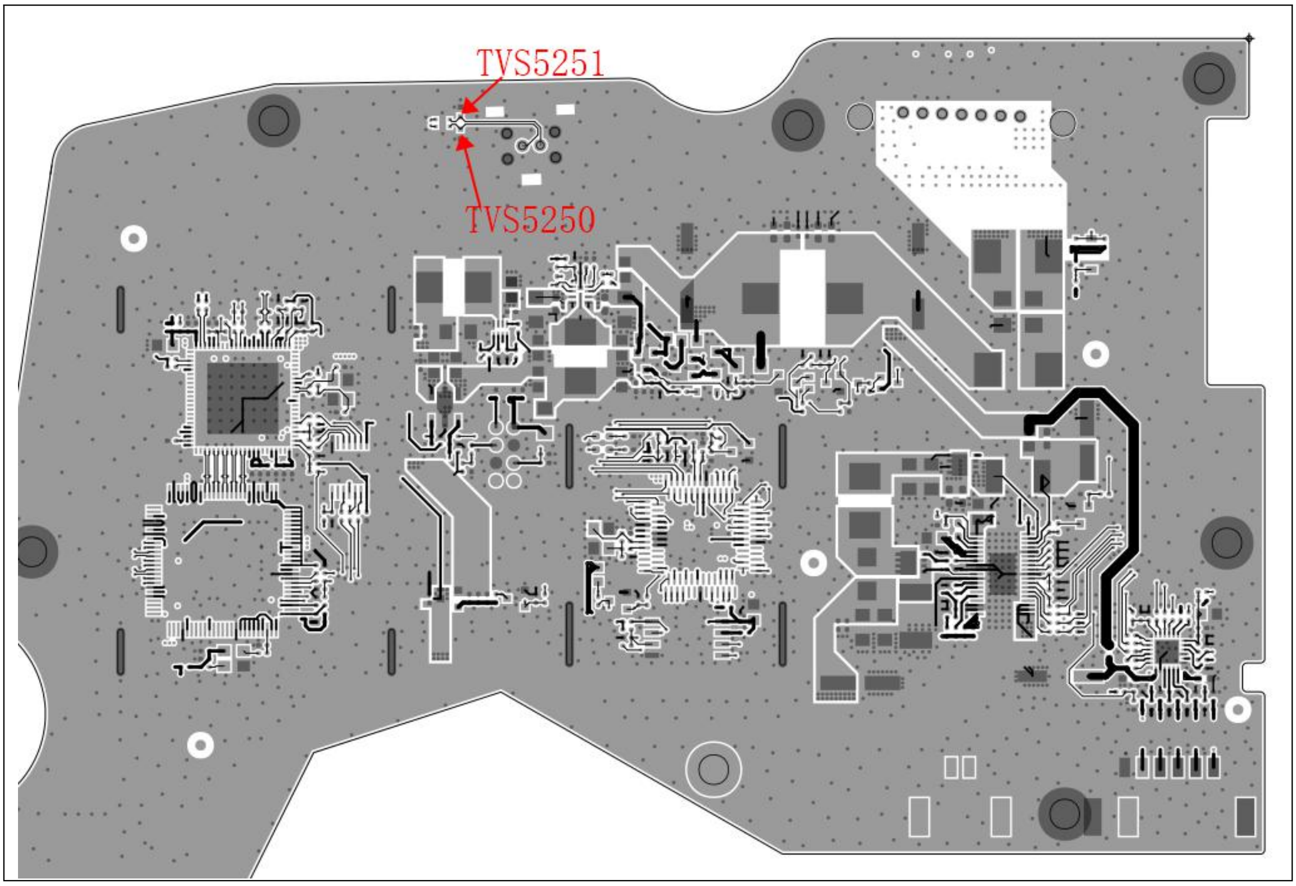
TVS8000 for VBAT:





TVS5250 for RIN0+ \ TVS5251 for RIN0- :





3. Eye Diagram Test

Test object	Inspection method	Test Site	Test Requirements	Remark
PGU LVDS lines	Eye Diagram Test	Third-party Lab	An eye diagram template is formed according to the parameters in the chip's specifications, and the tested eye diagram needs to meet the template requirements.	Provide the Eye Diagram Test report.

————The End————